

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
MTD selection (%)						
U2 - shared alpha	68	13.6	4.7	1.7	0	12
U2 - dose-specific alpha	67.7	13.6	5	1.7	0	12
U2 - cycle-1 oracle	65.6	14.7	7	2.6	0.6	9.5
U3 - shared alpha	68.3	13	4.9	1.7	0	12.1
U3 - dose-specific alpha	68.1	13.4	4.8	1.6	0.1	12
U3 - cycle-1 oracle	65.3	14.7	6.7	3.1	0.7	9.5
OBD selection (%)						
U2 - shared alpha	14.7	7.2	1	0.2	0	76.9
U2 - dose-specific alpha	14.8	6.9	1.4	0.1	0	76.8
U2 - cycle-1 oracle	16.6	6.7	2	0.4	0.2	74.1
U3 - shared alpha	15.8	6.2	1.1	0.1	0	76.8
U3 - dose-specific alpha	16.1	5.6	1	0.2	0	77.1
U3 - cycle-1 oracle	17	6.4	1.7	0.4	0.1	74.4
Mean administrations						
U2 - shared alpha	11.3	3.3	1.7	0.6	0.2	-
U2 - dose-specific alpha	11.3	3.3	1.7	0.6	0.2	-
U2 - cycle-1 oracle	11.1	3.6	2.1	0.9	0.3	-
U3 - shared alpha	11.3	3.3	1.7	0.6	0.2	-
U3 - dose-specific alpha	11.3	3.4	1.7	0.6	0.2	-
U3 - cycle-1 oracle	11	3.6	2.1	0.9	0.3	-
Mean IPDE administrations						
U2 - shared alpha	3.3	1.7	0.6	0.3	0.1	-
U2 - dose-specific alpha	3.3	1.7	0.6	0.3	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	3.3	1.7	0.6	0.3	0.1	-
U3 - dose-specific alpha	3.3	1.7	0.6	0.3	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						607.9
U2 - dose-specific alpha						607.7
U2 - cycle-1 oracle						985.6
U3 - shared alpha						608.2
U3 - dose-specific alpha						608.4
U3 - cycle-1 oracle						986.4
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
MTD selection (%)						
U2 - shared alpha	24.7	20.9	30.2	17.1	6.6	0.5
U2 - dose-specific alpha	24.6	21.2	30	17.1	6.6	0.5
U2 - cycle-1 oracle	22.6	19.1	30	18.5	9.3	0.5
U3 - shared alpha	24.5	21.5	29.1	18.2	6.2	0.5
U3 - dose-specific alpha	24.7	20.8	30.6	17	6.4	0.5
U3 - cycle-1 oracle	22.7	18.7	30.3	18.9	8.9	0.5
OBD selection (%)						
U2 - shared alpha	34.3	17.3	31.6	9.9	2.1	4.8
U2 - dose-specific alpha	33.9	17.9	31.5	9.4	2.3	5
U2 - cycle-1 oracle	33.4	16.4	31.6	10.3	3.2	5.1
U3 - shared alpha	36.9	18.6	29.4	8.9	1.3	4.9
U3 - dose-specific alpha	37.2	18.4	30.2	8	1.4	4.8
U3 - cycle-1 oracle	35.6	16.8	30.6	9.9	1.8	5.3
Mean administrations						
U2 - shared alpha	11.5	5.3	7.1	3.6	1.6	-
U2 - dose-specific alpha	11.5	5.4	7	3.6	1.6	-
U2 - cycle-1 oracle	10.9	5.4	7.1	3.9	1.9	-
U3 - shared alpha	11.6	5.3	7.1	3.7	1.6	-
U3 - dose-specific alpha	11.5	5.3	7.1	3.6	1.6	-
U3 - cycle-1 oracle	10.9	5.3	7.2	3.9	1.9	-
Mean IPDE administrations						
U2 - shared alpha	3.3	2.5	2	1	0.4	-
U2 - dose-specific alpha	3.3	2.5	1.9	1	0.4	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	3.4	2.5	1.9	1	0.4	-
U3 - dose-specific alpha	3.3	2.5	2	1	0.4	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1098.9
U2 - dose-specific alpha						1098.1
U2 - cycle-1 oracle						1617.8
U3 - shared alpha						1099.3
U3 - dose-specific alpha						1099
U3 - cycle-1 oracle						1617.6

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
MTD selection (%)						
U2 - shared alpha	25.7	20.9	28.9	17.5	6.4	0.6
U2 - dose-specific alpha	25.6	21	29.2	17.5	6.1	0.6
U2 - cycle-1 oracle	23.5	17.8	30.3	18.2	9.7	0.5
U3 - shared alpha	26	20.4	29.4	17.4	6.2	0.6
U3 - dose-specific alpha	25.5	21	30	16.3	6.6	0.6
U3 - cycle-1 oracle	23.4	17.8	30	19.2	9.1	0.5
OBD selection (%)						
U2 - shared alpha	12.7	41.7	18.3	6.9	1.9	18.5
U2 - dose-specific alpha	12.6	41.1	18.6	7.8	1.5	18.4
U2 - cycle-1 oracle	11.1	40.1	20.3	7.2	2.5	18.8
U3 - shared alpha	13.7	43.3	17	6.1	1.3	18.6
U3 - dose-specific alpha	13	43.9	16.9	6.2	1.5	18.5
U3 - cycle-1 oracle	12	42	19.2	6.2	1.8	18.8
Mean administrations						
U2 - shared alpha	8.6	7.7	5.9	3.3	1.5	-
U2 - dose-specific alpha	8.6	7.7	5.9	3.4	1.5	-
U2 - cycle-1 oracle	7.8	7.7	6.1	3.6	1.7	-
U3 - shared alpha	8.6	7.7	5.8	3.4	1.5	-
U3 - dose-specific alpha	8.6	7.7	6	3.3	1.5	-
U3 - cycle-1 oracle	7.8	7.6	6.2	3.6	1.7	-
Mean IPDE administrations						
U2 - shared alpha	2.3	3.3	1.7	1.1	0.4	-
U2 - dose-specific alpha	2.3	3.3	1.6	1.1	0.4	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.3	3.3	1.6	1.1	0.4	-
U3 - dose-specific alpha	2.3	3.3	1.7	1	0.5	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1004.8
U2 - dose-specific alpha						1005.3
U2 - cycle-1 oracle						1500.2
U3 - shared alpha						1005
U3 - dose-specific alpha						1005.1
U3 - cycle-1 oracle						1499.8
Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
MTD selection (%)						
U2 - shared alpha	2.7	3.2	24	42.7	27.4	0
U2 - dose-specific alpha	2.7	3.5	23.5	43	27.3	0
U2 - cycle-1 oracle	2.1	3	21.7	40.1	33.1	0
U3 - shared alpha	2.7	3.4	24.6	41.8	27.5	0
U3 - dose-specific alpha	2.7	3.4	24	42.6	27.3	0
U3 - cycle-1 oracle	2.1	2.7	21.8	40.5	32.9	0
OBD selection (%)						
U2 - shared alpha	7.2	7.6	23.4	42.8	15.4	3.6
U2 - dose-specific alpha	7.2	8.5	22.8	43.6	14.3	3.6
U2 - cycle-1 oracle	7	6.8	21.9	44.5	16.9	2.9
U3 - shared alpha	11.1	9.2	21.4	41.2	13.5	3.6
U3 - dose-specific alpha	11.3	9.1	21.8	42	12.3	3.5
U3 - cycle-1 oracle	11.2	8.4	21.1	42.1	14.3	2.9
Mean administrations						
U2 - shared alpha	4.4	4.3	6.8	8.1	5.8	-
U2 - dose-specific alpha	4.4	4.3	6.7	8.2	5.8	-
U2 - cycle-1 oracle	4.3	4.3	6.7	8.1	6	-
U3 - shared alpha	4.4	4.3	6.7	8.1	5.8	-
U3 - dose-specific alpha	4.4	4.4	6.7	8.1	5.8	-
U3 - cycle-1 oracle	4.3	4.2	6.7	8.1	6	-
Mean IPDE administrations						
U2 - shared alpha	0.6	2.8	2.1	2.7	1.4	-
U2 - dose-specific alpha	0.6	2.8	2.1	2.7	1.4	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.7	2.8	2.1	2.7	1.4	-
U3 - dose-specific alpha	0.6	2.8	2.1	2.6	1.4	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1089.2
U2 - dose-specific alpha						1089.3
U2 - cycle-1 oracle						1630.5
U3 - shared alpha						1088.5
U3 - dose-specific alpha						1090.2
U3 - cycle-1 oracle						1629.2

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
MTD selection (%)						
U2 - shared alpha	6.2	3.9	14.4	26.7	48.7	0.1
U2 - dose-specific alpha	6.2	3.9	14.3	27.1	48.4	0.1
U2 - cycle-1 oracle	5.6	3.3	13.4	23.3	54.3	0.1
U3 - shared alpha	6.2	4.1	14	26.9	48.7	0.1
U3 - dose-specific alpha	6.2	3.9	14.8	25.6	49.4	0.1
U3 - cycle-1 oracle	5.6	3.2	13.4	23.6	54.1	0.1
OBD selection (%)						
U2 - shared alpha	3.9	36.4	25.8	18.4	9.4	6.1
U2 - dose-specific alpha	3.2	35.4	25.4	20	9.9	6.1
U2 - cycle-1 oracle	3.4	37.6	25.5	18.8	9.2	5.5
U3 - shared alpha	6.8	41.1	24.5	16.3	5.2	6.1
U3 - dose-specific alpha	6.2	41.2	25	15.9	5.6	6.1
U3 - cycle-1 oracle	6.5	44.3	22	15.1	6.6	5.5
Mean administrations						
U2 - shared alpha	4.6	6.6	6.7	5.8	5.2	-
U2 - dose-specific alpha	4.6	6.6	6.7	5.8	5.2	-
U2 - cycle-1 oracle	4.4	6.7	6.5	5.9	5.3	-
U3 - shared alpha	4.5	6.6	6.7	5.8	5.1	-
U3 - dose-specific alpha	4.5	6.5	6.8	5.8	5.2	-
U3 - cycle-1 oracle	4.4	6.6	6.6	5.9	5.4	-
Mean IPDE administrations						
U2 - shared alpha	0.7	3.5	2.1	2.2	1.8	-
U2 - dose-specific alpha	0.7	3.5	2.1	2.2	1.8	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.7	3.5	2.2	2.2	1.8	-
U3 - dose-specific alpha	0.7	3.5	2.2	2.2	1.8	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1014.1
U2 - dose-specific alpha						1014.5
U2 - cycle-1 oracle						1603.3
U3 - shared alpha						1014.8
U3 - dose-specific alpha						1015.3
U3 - cycle-1 oracle						1603.3
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
MTD selection (%)						
U2 - shared alpha	59.5	16	7.2	2.4	0.3	14.6
U2 - dose-specific alpha	59.1	16.7	6.3	2.7	0.4	14.8
U2 - cycle-1 oracle	56.4	17.5	9	3.6	0.7	12.8
U3 - shared alpha	59.5	16.3	6.7	2.4	0.4	14.7
U3 - dose-specific alpha	59.6	16.3	6.4	2.7	0.3	14.7
U3 - cycle-1 oracle	56.1	17.9	8.6	3.7	0.7	13
OBD selection (%)						
U2 - shared alpha	66.6	11.3	3	0.6	0.2	18.3
U2 - dose-specific alpha	66.7	11.5	2.8	0.8	0.3	17.9
U2 - cycle-1 oracle	65.3	13	3.7	1	0.4	16.6
U3 - shared alpha	67.9	11.2	2.5	0.2	0.2	18
U3 - dose-specific alpha	67.2	11.7	2.6	0.5	0.1	17.9
U3 - cycle-1 oracle	66.8	12.2	3.3	1	0.3	16.4
Mean administrations						
U2 - shared alpha	19.6	3.8	2.2	0.8	0.2	-
U2 - dose-specific alpha	19.7	3.8	2.2	0.8	0.3	-
U2 - cycle-1 oracle	19.1	4	2.5	1.1	0.4	-
U3 - shared alpha	19.7	3.8	2.1	0.9	0.3	-
U3 - dose-specific alpha	19.7	3.9	2.1	0.9	0.3	-
U3 - cycle-1 oracle	19.1	4	2.5	1.1	0.3	-
Mean IPDE administrations						
U2 - shared alpha	5.4	1.5	0.6	0.3	0.1	-
U2 - dose-specific alpha	5.4	1.5	0.6	0.3	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	5.4	1.5	0.6	0.3	0.1	-
U3 - dose-specific alpha	5.4	1.5	0.6	0.3	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1034.5
U2 - dose-specific alpha						1037.8
U2 - cycle-1 oracle						1501.6
U3 - shared alpha						1037
U3 - dose-specific alpha						1038.3
U3 - cycle-1 oracle						1502.5

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
MTD selection (%)						
U2 - shared alpha	26	16.8	22.5	22.4	11.6	0.7
U2 - dose-specific alpha	26	16.6	22.9	22	11.8	0.7
U2 - cycle-1 oracle	23.4	15.2	23.3	21.6	15.8	0.7
U3 - shared alpha	26	17.1	23	21.1	12.1	0.7
U3 - dose-specific alpha	26	16.6	23.3	21.6	11.8	0.7
U3 - cycle-1 oracle	23.3	15.4	23.7	20.5	16.4	0.7
OBD selection (%)						
U2 - shared alpha	25.2	36.8	19.6	10.6	2.2	5.6
U2 - dose-specific alpha	25.1	38	19.6	9.6	2.1	5.6
U2 - cycle-1 oracle	24.3	36.3	20.3	9.8	3.4	5.9
U3 - shared alpha	27.4	39.8	17	8.8	1.4	5.6
U3 - dose-specific alpha	26.8	41.1	16.6	8.4	1.6	5.5
U3 - cycle-1 oracle	25.4	41.1	17	8.5	2.1	5.9
Mean administrations						
U2 - shared alpha	10.9	7	5.5	3.7	1.9	-
U2 - dose-specific alpha	10.9	7	5.5	3.6	2	-
U2 - cycle-1 oracle	10.2	7.2	5.6	3.7	2.3	-
U3 - shared alpha	10.8	7.1	5.5	3.6	2	-
U3 - dose-specific alpha	10.9	7.1	5.5	3.6	1.9	-
U3 - cycle-1 oracle	10.1	7.3	5.6	3.8	2.3	-
Mean IPDE administrations						
U2 - shared alpha	3	2.8	1.3	1	0.5	-
U2 - dose-specific alpha	3.1	2.8	1.3	1	0.5	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	3	2.8	1.3	1	0.5	-
U3 - dose-specific alpha	3	2.9	1.4	1	0.5	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1114.9
U2 - dose-specific alpha						1113.9
U2 - cycle-1 oracle						1608.4
U3 - shared alpha						1114.4
U3 - dose-specific alpha						1114.3
U3 - cycle-1 oracle						1608.4
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
MTD selection (%)						
U2 - shared alpha	67.6	9.2	4	1	0.2	18
U2 - dose-specific alpha	66.5	9.5	4.1	1.2	0.1	18.6
U2 - cycle-1 oracle	66.9	11.8	5	1.8	0.5	14
U3 - shared alpha	67.6	9.3	4	0.9	0.2	18
U3 - dose-specific alpha	67.4	9.1	4.1	1.1	0.2	18.1
U3 - cycle-1 oracle	66.1	12.1	5.4	1.6	0.5	14.3
OBD selection (%)						
U2 - shared alpha	28.7	5.7	1.9	0.7	0.2	62.8
U2 - dose-specific alpha	27.8	6.5	1.8	0.9	0.1	62.9
U2 - cycle-1 oracle	25.9	9	2.2	0.9	0.3	61.7
U3 - shared alpha	29.8	5.4	1.2	0.5	0.2	62.9
U3 - dose-specific alpha	29.4	5.1	1.5	0.7	0.2	63.1
U3 - cycle-1 oracle	27	8.5	1.8	1	0.4	61.3
Mean administrations						
U2 - shared alpha	14	3.3	1.5	0.4	0.1	-
U2 - dose-specific alpha	14	3.3	1.5	0.5	0.1	-
U2 - cycle-1 oracle	13.3	3.7	1.9	0.7	0.2	-
U3 - shared alpha	14	3.3	1.5	0.4	0.1	-
U3 - dose-specific alpha	14	3.3	1.5	0.5	0.1	-
U3 - cycle-1 oracle	13.4	3.6	1.9	0.8	0.2	-
Mean IPDE administrations						
U2 - shared alpha	4.2	1.6	0.5	0.2	0	-
U2 - dose-specific alpha	4.1	1.6	0.5	0.2	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	4.2	1.6	0.5	0.2	0	-
U3 - dose-specific alpha	4.2	1.5	0.5	0.2	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						705.9
U2 - dose-specific alpha						704.7
U2 - cycle-1 oracle						1094.9
U3 - shared alpha						704.7
U3 - dose-specific alpha						705.8
U3 - cycle-1 oracle						1096.2

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
MTD selection (%)						
U2 - shared alpha	25.4	11.5	16.2	18.7	27.6	0.6
U2 - dose-specific alpha	25.5	11.3	16.5	18.7	27.4	0.6
U2 - cycle-1 oracle	23.1	9	17.3	18.4	31.7	0.5
U3 - shared alpha	25.3	11.6	16.9	17.1	28.5	0.6
U3 - dose-specific alpha	25.3	11.8	16.5	17.5	28.3	0.6
U3 - cycle-1 oracle	23.1	8.9	17.9	17.9	31.7	0.5
OBD selection (%)						
U2 - shared alpha	9.3	9.8	16.8	12.4	22.2	29.5
U2 - dose-specific alpha	8.7	10.9	17.2	11.9	21.9	29.4
U2 - cycle-1 oracle	9.2	10	16.9	13.7	23.7	26.5
U3 - shared alpha	13.6	10.8	15.5	10	21	29.1
U3 - dose-specific alpha	14.3	10.2	15.3	10	20.6	29.6
U3 - cycle-1 oracle	13.9	10.9	16.9	11	20.5	26.8
Mean administrations						
U2 - shared alpha	7.3	4.3	5.3	3.7	4.5	-
U2 - dose-specific alpha	7.2	4.3	5.4	3.7	4.5	-
U2 - cycle-1 oracle	6.9	4.3	5.5	4.1	4.8	-
U3 - shared alpha	7.2	4.3	5.4	3.6	4.6	-
U3 - dose-specific alpha	7.3	4.2	5.4	3.7	4.5	-
U3 - cycle-1 oracle	6.9	4.2	5.6	4	4.8	-
Mean IPDE administrations						
U2 - shared alpha	1.9	2.5	1.8	1.5	1.3	-
U2 - dose-specific alpha	1.9	2.5	1.9	1.5	1.3	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.9	2.5	1.9	1.5	1.3	-
U3 - dose-specific alpha	1.9	2.4	1.9	1.5	1.3	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						882.4
U2 - dose-specific alpha						882.2
U2 - cycle-1 oracle						1423.5
U3 - shared alpha						882.7
U3 - dose-specific alpha						881.1
U3 - cycle-1 oracle						1423.8
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
MTD selection (%)						
U2 - shared alpha	53	18.2	14.6	7.5	2.7	4
U2 - dose-specific alpha	52.7	18	15.3	7.3	2.6	4.1
U2 - cycle-1 oracle	47.9	19	15.4	9.8	4.6	3.3
U3 - shared alpha	53.2	17.8	15	7.3	2.7	4
U3 - dose-specific alpha	52.9	18	15.2	7.3	2.6	4
U3 - cycle-1 oracle	47.8	18.9	15.7	9.8	4.5	3.3
OBD selection (%)						
U2 - shared alpha	7.8	25.1	10.8	3.9	1.2	51.2
U2 - dose-specific alpha	8	25.5	10.7	3.5	1.1	51.2
U2 - cycle-1 oracle	7.1	28.5	11.9	3.6	2	46.9
U3 - shared alpha	8.1	26.4	10	2.9	1.1	51.5
U3 - dose-specific alpha	8.5	27.7	9	2.9	0.9	51
U3 - cycle-1 oracle	7.4	30.3	10.6	3.5	1.4	46.8
Mean administrations						
U2 - shared alpha	9.2	5.5	3.7	1.8	0.8	-
U2 - dose-specific alpha	9.2	5.5	3.7	1.8	0.7	-
U2 - cycle-1 oracle	8.4	6	4.1	2.1	1	-
U3 - shared alpha	9.2	5.5	3.7	1.8	0.7	-
U3 - dose-specific alpha	9.2	5.5	3.7	1.8	0.8	-
U3 - cycle-1 oracle	8.4	6	4.1	2.1	1	-
Mean IPDE administrations						
U2 - shared alpha	2.5	2.4	1	0.6	0.2	-
U2 - dose-specific alpha	2.5	2.5	1	0.6	0.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.5	2.4	1	0.6	0.2	-
U3 - dose-specific alpha	2.5	2.4	1	0.6	0.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						776.2
U2 - dose-specific alpha						776.6
U2 - cycle-1 oracle						1195.7
U3 - shared alpha						776.4
U3 - dose-specific alpha						776
U3 - cycle-1 oracle						1195

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
MTD selection (%)						
U2 - shared alpha	53.9	30.6	10.8	0.5	0	4.2
U2 - dose-specific alpha	54.3	30.8	10.3	0.6	0	4
U2 - cycle-1 oracle	49.4	31.8	13.3	1.9	0	3.6
U3 - shared alpha	54.9	29.2	11.3	0.4	0	4.2
U3 - dose-specific alpha	54.1	30.8	10.6	0.3	0	4.2
U3 - cycle-1 oracle	48.8	32.6	12.9	2.1	0	3.6
OBD selection (%)						
U2 - shared alpha	63.8	18.9	2.6	0.2	0	14.5
U2 - dose-specific alpha	63.9	18.9	2.7	0.1	0	14.4
U2 - cycle-1 oracle	58.9	20.5	4	0.6	0	16
U3 - shared alpha	65.6	17.8	2	0.1	0	14.5
U3 - dose-specific alpha	64.9	18.2	2.2	0.1	0	14.6
U3 - cycle-1 oracle	60.8	19.8	2.9	0.6	0	15.9
Mean administrations						
U2 - shared alpha	18.1	5.7	2.9	0.7	0.1	-
U2 - dose-specific alpha	18.1	5.7	2.8	0.7	0.1	-
U2 - cycle-1 oracle	17	5.8	3.5	1	0.1	-
U3 - shared alpha	18.1	5.7	2.9	0.7	0.1	-
U3 - dose-specific alpha	18.1	5.7	2.8	0.7	0.1	-
U3 - cycle-1 oracle	16.9	5.9	3.5	0.9	0.1	-
Mean IPDE administrations						
U2 - shared alpha	5.6	2.5	0.9	0.3	0	-
U2 - dose-specific alpha	5.6	2.5	0.9	0.3	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	5.6	2.5	0.9	0.3	0	-
U3 - dose-specific alpha	5.6	2.5	0.9	0.3	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						998.9
U2 - cycle-1 oracle						998
U3 - shared alpha						1520.9
U3 - dose-specific alpha						997.9
U3 - cycle-1 oracle						998.1
						1521.4
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
MTD selection (%)						
U2 - shared alpha	4.4	2.7	24.7	45.2	22.9	0.1
U2 - dose-specific alpha	4.4	2.7	25.2	44.4	23.2	0.1
U2 - cycle-1 oracle	4	2.7	21.9	41.7	29.6	0.1
U3 - shared alpha	4.4	2.6	25.1	44.5	23.3	0.1
U3 - dose-specific alpha	4.4	2.6	24.9	44.7	23.3	0.1
U3 - cycle-1 oracle	3.9	2.6	21.9	43	28.5	0.1
OBD selection (%)						
U2 - shared alpha	13.8	13.9	11.2	38.9	16.8	5.4
U2 - dose-specific alpha	14.1	14.8	11.5	36.5	17.4	5.7
U2 - cycle-1 oracle	14	12.9	9.5	35.2	22.7	5.7
U3 - shared alpha	19.7	13.5	10.3	35.2	15.8	5.5
U3 - dose-specific alpha	20.6	14.2	9.3	35.5	15.1	5.3
U3 - cycle-1 oracle	19.2	14.2	7.2	33.7	20.1	5.6
Mean administrations						
U2 - shared alpha	4.9	4.7	5.6	7.4	6.5	-
U2 - dose-specific alpha	4.9	4.8	5.6	7.2	6.5	-
U2 - cycle-1 oracle	4.7	4.7	5.4	7.6	6.8	-
U3 - shared alpha	4.9	4.8	5.6	7.3	6.5	-
U3 - dose-specific alpha	4.9	4.8	5.6	7.3	6.5	-
U3 - cycle-1 oracle	4.8	4.7	5.4	7.5	6.8	-
Mean IPDE administrations						
U2 - shared alpha	0.9	2.9	1.8	2.8	1.4	-
U2 - dose-specific alpha	0.9	3	1.8	2.7	1.5	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.9	3	1.8	2.7	1.5	-
U3 - dose-specific alpha	0.9	3	1.8	2.8	1.4	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						1059.8
U2 - cycle-1 oracle						1059.5
U3 - shared alpha						1614.4
U3 - dose-specific alpha						1058.9
U3 - cycle-1 oracle						1060.6
						1615.1

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
MTD selection (%)						
U2 - shared alpha	48.8	20.3	17.6	8.1	2.4	2.8
U2 - dose-specific alpha	48.8	20	17.7	8.4	2.3	2.8
U2 - cycle-1 oracle	44	20.1	18.7	10.7	4.3	2.2
U3 - shared alpha	48.5	20.6	17.1	8.6	2.4	2.8
U3 - dose-specific alpha	48.5	20.5	17.5	8.5	2.2	2.8
U3 - cycle-1 oracle	43.9	20	19.2	10.1	4.6	2.2
OBD selection (%)						
U2 - shared alpha	9.3	28.1	11.1	4	1.2	46.3
U2 - dose-specific alpha	9.6	28	11.2	3.6	1.2	46.4
U2 - cycle-1 oracle	8.3	30.2	13.2	4.2	1.9	42.2
U3 - shared alpha	11	28.8	9.9	3.1	1.1	46.1
U3 - dose-specific alpha	10.4	29	9.6	3.8	1.1	46.1
U3 - cycle-1 oracle	10.2	30.9	11.5	3.8	1.5	42.1
Mean administrations						
U2 - shared alpha	9	6	4	1.9	0.8	-
U2 - dose-specific alpha	9	6	3.9	2	0.8	-
U2 - cycle-1 oracle	8.2	6.4	4.4	2.4	1.1	-
U3 - shared alpha	9	5.9	4	2	0.8	-
U3 - dose-specific alpha	9	6	4	2	0.8	-
U3 - cycle-1 oracle	8.2	6.3	4.5	2.4	1.1	-
Mean IPDE administrations						
U2 - shared alpha	2.4	2.8	1.2	0.7	0.2	-
U2 - dose-specific alpha	2.5	2.8	1.2	0.7	0.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.5	2.7	1.2	0.7	0.2	-
U3 - dose-specific alpha	2.5	2.8	1.2	0.7	0.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						788
U2 - dose-specific alpha						789.8
U2 - cycle-1 oracle						1239.9
U3 - shared alpha						788.7
U3 - dose-specific alpha						788
U3 - cycle-1 oracle						1239.7

Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
MTD selection (%)						
U2 - shared alpha	10.9	8.9	26.2	29.8	24	0.2
U2 - dose-specific alpha	11	8.8	26.1	29.9	24	0.2
U2 - cycle-1 oracle	9.4	7.1	24.6	31.7	27	0.2
U3 - shared alpha	10.9	8.8	25.9	30.9	23.3	0.2
U3 - dose-specific alpha	11.1	8.1	26.2	30.4	24	0.2
U3 - cycle-1 oracle	9.6	6.9	25.3	30.7	27.3	0.2
OBD selection (%)						
U2 - shared alpha	18.7	13.6	15.4	24.5	13.2	14.6
U2 - dose-specific alpha	18	14.8	14.3	26.1	12.6	14.2
U2 - cycle-1 oracle	18	12.7	14.5	27.8	14.2	12.8
U3 - shared alpha	25.2	15.6	11.1	23.9	9.7	14.5
U3 - dose-specific alpha	25.4	15.1	11.3	24.5	9.4	14.3
U3 - cycle-1 oracle	26.4	15.1	11.5	23.8	10.4	12.8
Mean administrations						
U2 - shared alpha	6.3	5.3	5.9	5.7	4.6	-
U2 - dose-specific alpha	6.2	5.4	5.8	5.7	4.6	-
U2 - cycle-1 oracle	5.9	5.4	5.9	6.1	4.8	-
U3 - shared alpha	6.2	5.3	6	5.7	4.6	-
U3 - dose-specific alpha	6.3	5.3	5.9	5.7	4.5	-
U3 - cycle-1 oracle	5.9	5.3	6	6.1	4.8	-
Mean IPDE administrations						
U2 - shared alpha	1.5	3.1	2	2.2	1.4	-
U2 - dose-specific alpha	1.4	3.1	2	2.2	1.4	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.4	3.1	2.1	2.2	1.4	-
U3 - dose-specific alpha	1.5	3.1	2.1	2.2	1.4	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						967.2
U2 - dose-specific alpha						968.8
U2 - cycle-1 oracle						1556.8
U3 - shared alpha						966.2
U3 - dose-specific alpha						964.8
U3 - cycle-1 oracle						1557.4

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
MTD selection (%)						
U2 - shared alpha	20.6	23	34.6	16.9	4.5	0.4
U2 - dose-specific alpha	20.7	23.3	34.6	16.6	4.4	0.4
U2 - cycle-1 oracle	18.4	20	35	17.3	8.9	0.4
U3 - shared alpha	20.7	23.3	34.9	16.1	4.6	0.4
U3 - dose-specific alpha	20.6	23	36	15.7	4.3	0.4
U3 - cycle-1 oracle	18.3	20.4	35.2	17.2	8.5	0.4
OBD selection (%)						
U2 - shared alpha	42.8	24.7	9.6	3.1	0.2	19.6
U2 - dose-specific alpha	42.8	25.1	9.2	3	0.3	19.6
U2 - cycle-1 oracle	41.5	24.6	10.6	3.9	0.8	18.6
U3 - shared alpha	50.2	21.9	6.5	1.5	0.3	19.6
U3 - dose-specific alpha	50.7	22.8	5.7	1.4	0.1	19.3
U3 - cycle-1 oracle	49.9	23.2	6.5	1.4	0.3	18.7
Mean administrations						
U2 - shared alpha	10.6	7.1	5.7	2.9	1.1	-
U2 - dose-specific alpha	10.5	7	5.7	3	1.2	-
U2 - cycle-1 oracle	10	6.7	5.9	3.3	1.4	-
U3 - shared alpha	10.5	7	5.7	3	1.2	-
U3 - dose-specific alpha	10.6	6.9	5.7	2.9	1.2	-
U3 - cycle-1 oracle	10.1	6.8	5.9	3.2	1.5	-
Mean IPDE administrations						
U2 - shared alpha	3.2	3.6	2	1.3	0.5	-
U2 - dose-specific alpha	3.2	3.5	2	1.3	0.5	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	3.2	3.5	2	1.3	0.5	-
U3 - dose-specific alpha	3.2	3.5	2	1.3	0.5	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - shared alpha						929.1
U2 - dose-specific alpha						928.5
U2 - cycle-1 oracle						1523.1
U3 - shared alpha						928.9
U3 - dose-specific alpha						929.4
U3 - cycle-1 oracle						1523.5
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
MTD selection (%)						
U2 - shared alpha	0.7	0.6	2.5	10.5	85.7	0
U2 - dose-specific alpha	0.7	0.6	2.8	10.2	85.7	0
U2 - cycle-1 oracle	0.7	0.6	2.7	10.3	85.7	0
U3 - shared alpha	0.7	0.6	2.7	10.8	85.2	0
U3 - dose-specific alpha	0.7	0.6	2.7	10.2	85.8	0
U3 - cycle-1 oracle	0.7	0.6	2.5	10.1	86.1	0
OBD selection (%)						
U2 - shared alpha	29	26	16.2	15.7	13	0.1
U2 - dose-specific alpha	29.9	25	16.3	16.5	12.2	0.1
U2 - cycle-1 oracle	32.2	23.9	16	16	11.8	0.1
U3 - shared alpha	38.9	28.3	13.5	13	6.2	0.1
U3 - dose-specific alpha	36.7	29.2	13.2	13.9	6.9	0.1
U3 - cycle-1 oracle	39.7	25.4	14.2	13.6	7	0.1
Mean administrations						
U2 - shared alpha	5.8	5.7	5.5	5.8	7.2	-
U2 - dose-specific alpha	5.8	5.7	5.5	5.8	7.2	-
U2 - cycle-1 oracle	5.9	5.5	5.5	6	7	-
U3 - shared alpha	5.8	5.7	5.5	5.8	7.1	-
U3 - dose-specific alpha	5.8	5.8	5.5	5.8	7.2	-
U3 - cycle-1 oracle	5.9	5.5	5.5	6	7	-
Mean IPDE administrations						
U2 - shared alpha	1.2	3.2	1.9	2.6	2.7	-
U2 - dose-specific alpha	1.2	3.2	1.9	2.6	2.8	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.2	3.2	1.9	2.7	2.7	-
U3 - dose-specific alpha	1.2	3.2	1.9	2.6	2.7	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - shared alpha						1007.4
U2 - dose-specific alpha						1008
U2 - cycle-1 oracle						1662.9
U3 - shared alpha						1008
U3 - dose-specific alpha						1007.7
U3 - cycle-1 oracle						1662.9

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
MTD selection (%)						
U2 - shared alpha	18.2	42.7	34.1	4.8	0	0.2
U2 - dose-specific alpha	18	41.4	35.7	4.7	0	0.2
U2 - cycle-1 oracle	15.9	40.8	35.9	7	0.2	0.2
U3 - shared alpha	17.8	42.3	35.5	4.2	0	0.2
U3 - dose-specific alpha	17.4	41.8	35.6	5	0	0.2
U3 - cycle-1 oracle	15.9	40.5	36.2	7	0.2	0.2
OBD selection (%)						
U2 - shared alpha	31.2	51.8	14.6	1.5	0	0.9
U2 - dose-specific alpha	30.8	50.8	16.2	1.3	0	0.9
U2 - cycle-1 oracle	30.4	51.1	15.6	1.8	0	1.1
U3 - shared alpha	33.3	51.9	12.8	1.1	0	0.9
U3 - dose-specific alpha	33.3	52.2	12.6	1	0	0.9
U3 - cycle-1 oracle	33.3	50.7	13.7	1.3	0	1
Mean administrations						
U2 - shared alpha	10.6	10.1	6.8	2	0.3	-
U2 - dose-specific alpha	10.7	10	6.9	2	0.3	-
U2 - cycle-1 oracle	10.1	10	7.2	2.3	0.4	-
U3 - shared alpha	10.6	10.1	6.9	2	0.3	-
U3 - dose-specific alpha	10.7	10.1	6.9	1.9	0.3	-
U3 - cycle-1 oracle	10	10	7.2	2.3	0.4	-
Mean IPDE administrations						
U2 - shared alpha	2.7	3.5	1.5	0.6	0.1	-
U2 - dose-specific alpha	2.7	3.4	1.6	0.6	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.7	3.5	1.5	0.6	0.1	-
U3 - dose-specific alpha	2.7	3.5	1.6	0.5	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1187.6
U2 - dose-specific alpha						1186.5
U2 - cycle-1 oracle						1653.1
U3 - shared alpha						1187.2
U3 - dose-specific alpha						1187.1
U3 - cycle-1 oracle						1653.1
Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
MTD selection (%)						
U2 - shared alpha	0.7	0.2	4.6	26	68.5	0
U2 - dose-specific alpha	0.7	0.2	4.7	26.4	68	0
U2 - cycle-1 oracle	0.7	0.1	3.7	25.7	69.8	0
U3 - shared alpha	0.7	0.3	4.5	27.6	66.9	0
U3 - dose-specific alpha	0.7	0.2	4.7	26.6	67.8	0
U3 - cycle-1 oracle	0.7	0.2	3.5	24.8	70.8	0
OBD selection (%)						
U2 - shared alpha	9.1	9.2	10.8	14	55.8	1.1
U2 - dose-specific alpha	10.4	10.1	9.6	14.4	54.6	0.9
U2 - cycle-1 oracle	10.7	8.6	10.4	15.1	54.3	0.9
U3 - shared alpha	16.1	12.7	9.4	12.6	48.1	1.1
U3 - dose-specific alpha	15.2	13.9	9.7	11.5	48.7	1
U3 - cycle-1 oracle	16.9	11.4	9.7	12.7	48.2	1.1
Mean administrations						
U2 - shared alpha	3.7	3.9	4.7	5.6	12	-
U2 - dose-specific alpha	3.7	3.8	4.8	5.5	12	-
U2 - cycle-1 oracle	3.7	3.9	4.8	5.9	11.6	-
U3 - shared alpha	3.7	3.9	4.7	5.6	12	-
U3 - dose-specific alpha	3.7	3.9	4.7	5.5	12.1	-
U3 - cycle-1 oracle	3.7	3.9	4.8	5.9	11.6	-
Mean IPDE administrations						
U2 - shared alpha	0.3	2.7	1.5	2.5	3.3	-
U2 - dose-specific alpha	0.3	2.7	1.6	2.5	3.4	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.3	2.7	1.6	2.5	3.3	-
U3 - dose-specific alpha	0.3	2.7	1.6	2.5	3.3	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1070.3
U2 - dose-specific alpha						1070.4
U2 - cycle-1 oracle						1655.7
U3 - shared alpha						1070.3
U3 - dose-specific alpha						1071.4
U3 - cycle-1 oracle						1655.4

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
MTD selection (%)						
U2 - shared alpha	25.6	43.7	27.5	3	0	0.2
U2 - dose-specific alpha	25.9	43.5	27.7	2.7	0	0.2
U2 - cycle-1 oracle	22.1	41.5	30.9	5.2	0.1	0.2
U3 - shared alpha	25.4	43.6	28.2	2.6	0	0.2
U3 - dose-specific alpha	25.8	43.2	28.1	2.7	0	0.2
U3 - cycle-1 oracle	21.7	42.7	30.1	5.2	0.1	0.2
OBD selection (%)						
U2 - shared alpha	18.4	49.9	13.7	0.9	0	17.1
U2 - dose-specific alpha	18.7	49.9	13.6	0.9	0	16.9
U2 - cycle-1 oracle	15.8	51.2	15	1.8	0.1	16.1
U3 - shared alpha	20.3	50.7	11.2	0.6	0	17.2
U3 - dose-specific alpha	20.6	49.8	11.4	0.7	0	17.5
U3 - cycle-1 oracle	18.1	51.1	13.5	1.2	0	16.1
Mean administrations						
U2 - shared alpha	8.2	10.5	6.5	1.7	0.3	-
U2 - dose-specific alpha	8.1	10.5	6.4	1.7	0.3	-
U2 - cycle-1 oracle	7.6	10.4	6.9	2	0.3	-
U3 - shared alpha	8.1	10.6	6.4	1.7	0.3	-
U3 - dose-specific alpha	8.1	10.5	6.5	1.7	0.3	-
U3 - cycle-1 oracle	7.5	10.3	7	2.1	0.3	-
Mean IPDE administrations						
U2 - shared alpha	2.1	4.5	1.8	0.6	0.1	-
U2 - dose-specific alpha	2.1	4.5	1.8	0.6	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.1	4.6	1.8	0.6	0.1	-
U3 - dose-specific alpha	2.1	4.5	1.9	0.6	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						986.3
U2 - dose-specific alpha						986.3
U2 - cycle-1 oracle						1512.8
U3 - shared alpha						986.1
U3 - dose-specific alpha						986.7
U3 - cycle-1 oracle						1512.6

Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
MTD selection (%)						
U2 - shared alpha	7.2	15.2	47.1	25.7	4.7	0.1
U2 - dose-specific alpha	7.1	14.7	48.6	25.1	4.4	0.1
U2 - cycle-1 oracle	6.3	14.6	44.9	25.2	8.9	0.1
U3 - shared alpha	6.8	15.6	47	25.6	4.9	0.1
U3 - dose-specific alpha	7.2	15.1	46.9	26	4.7	0.1
U3 - cycle-1 oracle	6.2	14.6	44.4	25.6	9.1	0.1
OBD selection (%)						
U2 - shared alpha	9.8	45.8	28.8	8.1	0.7	6.8
U2 - dose-specific alpha	11	44.4	27.7	9.2	0.8	6.9
U2 - cycle-1 oracle	11.4	46.2	26.9	8	1.4	6.1
U3 - shared alpha	15.5	48.2	21.6	7.2	0.7	6.8
U3 - dose-specific alpha	16	49	20.7	7.2	0.3	6.8
U3 - cycle-1 oracle	17.4	48.9	20.4	6.3	1	6
Mean administrations						
U2 - shared alpha	5.3	9	8.5	4.1	1.8	-
U2 - dose-specific alpha	5.3	9.1	8.4	4.1	1.8	-
U2 - cycle-1 oracle	5.1	8.9	8.4	4.4	2	-
U3 - shared alpha	5.3	9.1	8.3	4.1	1.7	-
U3 - dose-specific alpha	5.3	9	8.4	4.1	1.8	-
U3 - cycle-1 oracle	5	9	8.4	4.4	2	-
Mean IPDE administrations						
U2 - shared alpha	1	4.4	2.6	1.5	0.6	-
U2 - dose-specific alpha	1	4.4	2.6	1.5	0.6	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.1	4.4	2.6	1.6	0.6	-
U3 - dose-specific alpha	1	4.4	2.6	1.5	0.6	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1015.6
U2 - dose-specific alpha						1014.2
U2 - cycle-1 oracle						1598.6
U3 - shared alpha						1014.8
U3 - dose-specific alpha						1014.6
U3 - cycle-1 oracle						1599

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
MTD selection (%)						
U2 - shared alpha	56.1	1.9	0	0	0	42
U2 - dose-specific alpha	56.3	1.7	0	0	0	42
U2 - cycle-1 oracle	59.4	2.9	0.1	0	0	37.6
U3 - shared alpha	55.7	1.9	0	0	0	42.4
U3 - dose-specific alpha	55	2.1	0	0	0	42.9
U3 - cycle-1 oracle	59	3	0.1	0	0	37.9
OBD selection (%)						
U2 - shared alpha	54.9	0.7	0	0	0	44.4
U2 - dose-specific alpha	54.4	1	0	0	0	44.6
U2 - cycle-1 oracle	59.7	1.4	0	0	0	38.9
U3 - shared alpha	54.5	0.8	0	0	0	44.7
U3 - dose-specific alpha	55	1	0	0	0	44
U3 - cycle-1 oracle	60.3	1.1	0	0	0	38.6
Mean administrations						
U2 - shared alpha	20.4	1.9	0.4	0	0	-
U2 - dose-specific alpha	20.4	1.9	0.4	0	0	-
U2 - cycle-1 oracle	20.9	2	0.6	0	0	-
U3 - shared alpha	20.4	1.9	0.4	0	0	-
U3 - dose-specific alpha	20.4	1.9	0.4	0	0	-
U3 - cycle-1 oracle	20.9	2.1	0.6	0	0	-
Mean IPDE administrations						
U2 - shared alpha	4.2	0.6	0.1	0	0	-
U2 - dose-specific alpha	4.1	0.6	0.1	0	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	4.1	0.6	0.1	0	0	-
U3 - dose-specific alpha	4.2	0.6	0.1	0	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						987.5
U2 - dose-specific alpha						988.1
U2 - cycle-1 oracle						1306.9
U3 - shared alpha						988
U3 - dose-specific alpha						989.9
U3 - cycle-1 oracle						1308.7
Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
MTD selection (%)						
U2 - shared alpha	20.2	52.9	25.2	1.6	0	0.1
U2 - dose-specific alpha	20.2	52.4	25.2	2.1	0	0.1
U2 - cycle-1 oracle	20.2	49.1	28.9	1.7	0	0.1
U3 - shared alpha	21.3	52	25.2	1.4	0	0.1
U3 - dose-specific alpha	20.5	54	24.1	1.3	0	0.1
U3 - cycle-1 oracle	19.8	49.6	28.6	1.9	0	0.1
OBD selection (%)						
U2 - shared alpha	23.3	54.3	10.5	1.3	0	10.6
U2 - dose-specific alpha	24.4	54.3	9.5	1.3	0	10.5
U2 - cycle-1 oracle	23.1	52.9	11.2	1.3	0	11.5
U3 - shared alpha	27.9	52.5	8.2	1.1	0	10.3
U3 - dose-specific alpha	27.9	53	7.8	1	0	10.3
U3 - cycle-1 oracle	26	52	9.4	1.3	0	11.3
Mean administrations						
U2 - shared alpha	8.1	12.6	5.9	1.3	0.2	-
U2 - dose-specific alpha	8.1	12.7	5.8	1.3	0.2	-
U2 - cycle-1 oracle	7.7	12.1	6.3	1.7	0.2	-
U3 - shared alpha	8	12.7	5.9	1.3	0.2	-
U3 - dose-specific alpha	8.1	12.6	5.9	1.3	0.2	-
U3 - cycle-1 oracle	7.8	12.1	6.3	1.6	0.2	-
Mean IPDE administrations						
U2 - shared alpha	2	5.4	1.8	0.5	0	-
U2 - dose-specific alpha	2	5.5	1.7	0.5	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2	5.5	1.7	0.5	0	-
U3 - dose-specific alpha	2	5.4	1.7	0.5	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1005.7
U2 - dose-specific alpha						1005.2
U2 - cycle-1 oracle						1554.8
U3 - shared alpha						1007.6
U3 - dose-specific alpha						1006.9
U3 - cycle-1 oracle						1554.6

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
MTD selection (%)						
U2 - shared alpha	77.8	4.5	0	0	0	17.7
U2 - dose-specific alpha	77.7	4.9	0	0	0	17.4
U2 - cycle-1 oracle	80.8	4.8	0.4	0	0	14
U3 - shared alpha	77.9	4.6	0	0	0	17.5
U3 - dose-specific alpha	78.2	4.6	0	0	0	17.2
U3 - cycle-1 oracle	81	5	0.6	0	0	13.4
OBD selection (%)						
U2 - shared alpha	6.3	3.1	0	0	0	90.6
U2 - dose-specific alpha	6.3	3.3	0	0	0	90.4
U2 - cycle-1 oracle	7.1	2.6	0.3	0	0	90
U3 - shared alpha	6.3	3.1	0	0	0	90.6
U3 - dose-specific alpha	6.3	3	0	0	0	90.7
U3 - cycle-1 oracle	6.9	2.5	0.4	0	0	90.2
Mean administrations						
U2 - shared alpha	11.7	2.1	0.4	0	0	-
U2 - dose-specific alpha	11.7	2.1	0.4	0	0	-
U2 - cycle-1 oracle	11.7	2.5	0.6	0	0	-
U3 - shared alpha	11.7	2.1	0.4	0	0	-
U3 - dose-specific alpha	11.7	2.1	0.4	0	0	-
U3 - cycle-1 oracle	11.7	2.5	0.6	0	0	-
Mean IPDE administrations						
U2 - shared alpha	3.4	1.1	0.1	0	0	-
U2 - dose-specific alpha	3.4	1.1	0.1	0	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	3.3	1.1	0.1	0	0	-
U3 - dose-specific alpha	3.4	1.1	0.1	0	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						517.6
U2 - dose-specific alpha						517.4
U2 - cycle-1 oracle						813.8
U3 - shared alpha						517.5
U3 - dose-specific alpha						517.7
U3 - cycle-1 oracle						813

Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
MTD selection (%)						
U2 - shared alpha	0	0.1	0.6	4.4	94.9	0
U2 - dose-specific alpha	0	0.1	0.6	4.2	95.1	0
U2 - cycle-1 oracle	0	0.1	0.7	4	95.2	0
U3 - shared alpha	0	0.1	0.6	4.5	94.8	0
U3 - dose-specific alpha	0	0.1	0.6	4.4	94.9	0
U3 - cycle-1 oracle	0	0.1	0.7	4.3	94.9	0
OBD selection (%)						
U2 - shared alpha	0.5	1.7	4.4	46.4	46.9	0.1
U2 - dose-specific alpha	1.1	1.9	4.3	46.8	45.7	0.2
U2 - cycle-1 oracle	1	1.6	5.9	45.2	46.2	0.1
U3 - shared alpha	2.1	2.6	6	48.2	41	0.1
U3 - dose-specific alpha	2.3	2.7	7.1	47	40.8	0.1
U3 - cycle-1 oracle	2.9	2.8	6.8	49.3	38	0.2
Mean administrations						
U2 - shared alpha	3.2	3.2	3.7	8.3	11.5	-
U2 - dose-specific alpha	3.2	3.2	3.7	8.4	11.5	-
U2 - cycle-1 oracle	3.3	3.3	3.8	8.6	11.1	-
U3 - shared alpha	3.2	3.2	3.7	8.4	11.5	-
U3 - dose-specific alpha	3.2	3.2	3.7	8.3	11.6	-
U3 - cycle-1 oracle	3.3	3.3	3.8	8.6	11.1	-
Mean IPDE administrations						
U2 - shared alpha	0.1	2.4	1.2	3.2	3.2	-
U2 - dose-specific alpha	0.1	2.4	1.2	3.2	3.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.1	2.4	1.2	3.3	3.1	-
U3 - dose-specific alpha	0.1	2.4	1.2	3.2	3.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1097.9
U2 - dose-specific alpha						1099.1
U2 - cycle-1 oracle						1663.8
U3 - shared alpha						1098.5
U3 - dose-specific alpha						1099.3
U3 - cycle-1 oracle						1663.4

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
MTD selection (%)						
U2 - shared alpha	12.7	35	45	7.1	0	0.2
U2 - dose-specific alpha	12.7	34.8	45.1	7.1	0.1	0.2
U2 - cycle-1 oracle	11.4	33.1	46.1	8.9	0.3	0.2
U3 - shared alpha	12.4	34.9	45.4	7.1	0	0.2
U3 - dose-specific alpha	12.4	35	45.3	7.1	0	0.2
U3 - cycle-1 oracle	11.6	33.8	45.2	8.8	0.4	0.2
OBD selection (%)						
U2 - shared alpha	33.3	32.2	28.4	4.1	0	2
U2 - dose-specific alpha	33.6	31.4	28.9	4.2	0	1.9
U2 - cycle-1 oracle	32.9	28.9	32	3.5	0.1	2.6
U3 - shared alpha	39.7	31.5	23.8	3.1	0	1.9
U3 - dose-specific alpha	40.9	31.2	23.1	2.9	0	1.9
U3 - cycle-1 oracle	38	29.1	26.9	3.4	0	2.6
Mean administrations						
U2 - shared alpha	9.4	8.5	9	2.3	0.4	-
U2 - dose-specific alpha	9.5	8.4	9	2.3	0.4	-
U2 - cycle-1 oracle	9.1	8.3	9.2	2.6	0.4	-
U3 - shared alpha	9.5	8.5	9	2.3	0.4	-
U3 - dose-specific alpha	9.4	8.5	9	2.4	0.4	-
U3 - cycle-1 oracle	9.1	8.1	9.3	2.6	0.4	-
Mean IPDE administrations						
U2 - shared alpha	2.6	3.7	2.5	0.8	0.1	-
U2 - dose-specific alpha	2.6	3.7	2.5	0.8	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.6	3.7	2.5	0.8	0.1	-
U3 - dose-specific alpha	2.6	3.7	2.5	0.8	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1105.9
U2 - dose-specific alpha						1104.9
U2 - cycle-1 oracle						1643.7
U3 - shared alpha						1105.5
U3 - dose-specific alpha						1105.3
U3 - cycle-1 oracle						1643.7
Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
MTD selection (%)						
U2 - shared alpha	28.4	30.2	33.6	7.1	0.1	0.6
U2 - dose-specific alpha	28.7	30.4	32.3	7.9	0.1	0.6
U2 - cycle-1 oracle	25.3	29	35.5	9.5	0.2	0.5
U3 - shared alpha	28.4	30.4	32.8	7.8	0	0.6
U3 - dose-specific alpha	28.9	29.5	33.5	7.5	0	0.6
U3 - cycle-1 oracle	25.8	28.1	35.3	10	0.3	0.5
OBD selection (%)						
U2 - shared alpha	22.2	29.6	25.2	3.1	0	19.9
U2 - dose-specific alpha	22	30	24.4	3.8	0	19.8
U2 - cycle-1 oracle	20.7	26.4	26.9	5.2	0	20.8
U3 - shared alpha	26	28	21.8	3.8	0	20.4
U3 - dose-specific alpha	26.7	26.9	23	3.1	0	20.3
U3 - cycle-1 oracle	24.1	26.2	23.2	5.3	0.1	21.1
Mean administrations						
U2 - shared alpha	9.9	6.9	7.5	2.3	0.3	-
U2 - dose-specific alpha	9.8	6.9	7.6	2.3	0.3	-
U2 - cycle-1 oracle	9	6.7	7.9	2.8	0.4	-
U3 - shared alpha	9.9	6.8	7.6	2.3	0.3	-
U3 - dose-specific alpha	9.9	7	7.5	2.3	0.3	-
U3 - cycle-1 oracle	9	6.6	8	2.8	0.4	-
Mean IPDE administrations						
U2 - shared alpha	2.9	3.3	2.2	0.8	0.1	-
U2 - dose-specific alpha	2.8	3.3	2.2	0.8	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.9	3.3	2.2	0.8	0.1	-
U3 - dose-specific alpha	2.9	3.3	2.2	0.8	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						973.6
U2 - dose-specific alpha						973.4
U2 - cycle-1 oracle						1488.3
U3 - shared alpha						971.7
U3 - dose-specific alpha						971.9
U3 - cycle-1 oracle						1487.7

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
MTD selection (%)						
U2 - shared alpha	15.8	25	43.2	15	0.8	0.2
U2 - dose-specific alpha	16.2	24.3	43.6	15	0.7	0.2
U2 - cycle-1 oracle	13.9	22.2	46.3	15.8	1.6	0.2
U3 - shared alpha	16	24.7	44.1	14.2	0.8	0.2
U3 - dose-specific alpha	16.1	25.1	43.6	14.3	0.7	0.2
U3 - cycle-1 oracle	13.9	21.4	47.1	15.7	1.7	0.2
OBD selection (%)						
U2 - shared alpha	18	25.1	37.4	6.9	0.3	12.3
U2 - dose-specific alpha	17.4	24.6	39	6.3	0.3	12.4
U2 - cycle-1 oracle	16.7	22.7	41.5	6.7	0.7	11.7
U3 - shared alpha	20.8	25.2	36.4	5.4	0.1	12.1
U3 - dose-specific alpha	21.6	25.9	34.4	5.8	0.1	12.2
U3 - cycle-1 oracle	19.7	22.8	38.2	7.1	0.4	11.8
Mean administrations						
U2 - shared alpha	7.6	6.7	9.1	3.6	1.1	-
U2 - dose-specific alpha	7.7	6.7	9.1	3.5	1.1	-
U2 - cycle-1 oracle	6.9	6.7	9.4	3.9	1.2	-
U3 - shared alpha	7.6	6.7	9.1	3.6	1.1	-
U3 - dose-specific alpha	7.7	6.7	9	3.6	1.1	-
U3 - cycle-1 oracle	6.9	6.6	9.5	3.9	1.2	-
Mean IPDE administrations						
U2 - shared alpha	2	3.3	2.6	1.1	0.3	-
U2 - dose-specific alpha	2	3.3	2.6	1.1	0.3	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2	3.3	2.6	1.1	0.3	-
U3 - dose-specific alpha	2	3.3	2.6	1.1	0.3	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1037.9
U2 - dose-specific alpha						1036.7
U2 - cycle-1 oracle						1564.2
U3 - shared alpha						1037.9
U3 - dose-specific alpha						1037
U3 - cycle-1 oracle						1563.9

Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
MTD selection (%)						
U2 - shared alpha	57.4	30.4	9.5	0.3	0	2.4
U2 - dose-specific alpha	56.8	31.1	9.4	0.1	0	2.6
U2 - cycle-1 oracle	53.2	32	11.6	0.9	0.1	2.2
U3 - shared alpha	57	30.9	9.4	0.1	0	2.6
U3 - dose-specific alpha	57.7	30.5	8.8	0.4	0	2.6
U3 - cycle-1 oracle	53.1	32	11.7	0.9	0.1	2.2
OBD selection (%)						
U2 - shared alpha	16.7	31.1	4.8	0.1	0	47.3
U2 - dose-specific alpha	16.6	30.9	5.1	0	0	47.4
U2 - cycle-1 oracle	16.1	34	5.4	0.4	0	44.1
U3 - shared alpha	17.3	31.9	3.6	0	0	47.2
U3 - dose-specific alpha	17.5	30.7	3.7	0.1	0	48
U3 - cycle-1 oracle	16	34.9	4.6	0.3	0	44.2
Mean administrations						
U2 - shared alpha	9.4	8.1	3.5	0.6	0.1	-
U2 - dose-specific alpha	9.5	8.1	3.5	0.6	0.1	-
U2 - cycle-1 oracle	8.8	8.5	4.1	0.8	0.1	-
U3 - shared alpha	9.4	8.2	3.5	0.6	0.1	-
U3 - dose-specific alpha	9.4	8.1	3.5	0.6	0.1	-
U3 - cycle-1 oracle	8.8	8.4	4.1	0.8	0.1	-
Mean IPDE administrations						
U2 - shared alpha	2.6	3.4	1	0.2	0	-
U2 - dose-specific alpha	2.6	3.4	1	0.2	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.6	3.4	1	0.2	0	-
U3 - dose-specific alpha	2.6	3.4	1	0.2	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						797.4
U2 - dose-specific alpha						797.9
U2 - cycle-1 oracle						1234.7
U3 - shared alpha						798.2
U3 - dose-specific alpha						796.9
U3 - cycle-1 oracle						1233.3

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
MTD selection (%)						
U2 - shared alpha	46.8	39.1	11	0.4	0	2.7
U2 - dose-specific alpha	47.1	38.5	11.2	0.5	0	2.7
U2 - cycle-1 oracle	45.1	39.1	12.5	0.8	0	2.5
U3 - shared alpha	47	38.9	10.8	0.6	0	2.7
U3 - dose-specific alpha	46.7	39.2	10.9	0.5	0	2.7
U3 - cycle-1 oracle	44.6	39.8	12.6	0.5	0	2.5
OBD selection (%)						
U2 - shared alpha	71	23	3	0.1	0	2.9
U2 - dose-specific alpha	70.8	23.1	2.9	0.2	0	3
U2 - cycle-1 oracle	68.9	23.8	4.1	0.4	0	2.8
U3 - shared alpha	72.8	21.7	2.3	0.2	0	3
U3 - dose-specific alpha	73.1	21.4	2.4	0.1	0	3
U3 - cycle-1 oracle	71.4	22.7	3.1	0.1	0	2.7
Mean administrations						
U2 - shared alpha	18.6	7.2	3.1	0.4	0	-
U2 - dose-specific alpha	18.7	7.2	3.1	0.4	0	-
U2 - cycle-1 oracle	18	7.4	3.5	0.5	0	-
U3 - shared alpha	18.7	7.2	3.1	0.4	0	-
U3 - dose-specific alpha	18.6	7.3	3.1	0.4	0	-
U3 - cycle-1 oracle	18.1	7.3	3.6	0.5	0	-
Mean IPDE administrations						
U2 - shared alpha	4.8	2.4	0.9	0.1	0	-
U2 - dose-specific alpha	4.9	2.4	0.9	0.1	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	4.8	2.4	0.8	0.1	0	-
U3 - dose-specific alpha	4.8	2.4	0.8	0.1	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1170.1
U2 - dose-specific alpha						1169.8
U2 - cycle-1 oracle						1634.8
U3 - shared alpha						1170.3
U3 - dose-specific alpha						1169.9
U3 - cycle-1 oracle						1635.1
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
MTD selection (%)						
U2 - shared alpha	2.6	2.3	26.2	47.7	21.2	0
U2 - dose-specific alpha	2.6	2.4	25.8	48.1	21.1	0
U2 - cycle-1 oracle	2.1	2.3	24.1	45.4	26.1	0
U3 - shared alpha	2.6	2.4	25.7	48	21.3	0
U3 - dose-specific alpha	2.6	2	26.9	47.6	20.9	0
U3 - cycle-1 oracle	2.1	2.1	24.4	46.1	25.3	0
OBD selection (%)						
U2 - shared alpha	18.8	15.5	15.6	37.6	10.9	1.6
U2 - dose-specific alpha	18.7	15.7	17.5	36.2	10.3	1.6
U2 - cycle-1 oracle	20.4	14	15.8	36.9	11.2	1.7
U3 - shared alpha	26.2	17.5	13.8	32.5	8.4	1.6
U3 - dose-specific alpha	25.5	17.6	13.8	32.2	9.2	1.7
U3 - cycle-1 oracle	27.9	16.7	12.9	32.2	8.6	1.7
Mean administrations						
U2 - shared alpha	5.2	4.6	6.2	8.1	5.6	-
U2 - dose-specific alpha	5.1	4.7	6.2	8	5.7	-
U2 - cycle-1 oracle	5	4.7	5.9	8.2	5.9	-
U3 - shared alpha	5.2	4.6	6.2	8.1	5.7	-
U3 - dose-specific alpha	5.2	4.7	6.1	8.1	5.6	-
U3 - cycle-1 oracle	4.9	4.7	5.9	8.2	5.8	-
Mean IPDE administrations						
U2 - shared alpha	1	2.8	2	2.8	1.4	-
U2 - dose-specific alpha	1	2.8	2	2.8	1.4	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1	2.8	2	2.8	1.4	-
U3 - dose-specific alpha	1	2.8	1.9	2.8	1.4	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1089.8
U2 - dose-specific alpha						1090
U2 - cycle-1 oracle						1646
U3 - shared alpha						1090.9
U3 - dose-specific alpha						1089.3
U3 - cycle-1 oracle						1646

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
MTD selection (%)						
U2 - shared alpha	1.6	1.3	10.1	43.6	43.4	0
U2 - dose-specific alpha	1.6	1.4	9.4	43.5	44.1	0
U2 - cycle-1 oracle	1.5	1.4	8.6	40.6	47.9	0
U3 - shared alpha	1.6	1.3	9.9	41.9	45.3	0
U3 - dose-specific alpha	1.6	1.4	9.5	43.2	44.3	0
U3 - cycle-1 oracle	1.5	1.4	8.1	41	48	0
OBD selection (%)						
U2 - shared alpha	6	34.7	27	24.5	7.5	0.3
U2 - dose-specific alpha	5.9	35.8	27.1	23.5	7.4	0.3
U2 - cycle-1 oracle	7.7	35.8	25.9	24	6.4	0.2
U3 - shared alpha	8.9	39.9	26	21	3.9	0.3
U3 - dose-specific alpha	8.5	43	24.9	18.9	4.4	0.3
U3 - cycle-1 oracle	10.1	42.2	24.1	19.5	3.9	0.2
Mean administrations						
U2 - shared alpha	4.1	5.7	6.4	7.9	5.8	-
U2 - dose-specific alpha	4.1	5.8	6.4	7.9	5.7	-
U2 - cycle-1 oracle	4.1	6.1	6.2	7.8	5.8	-
U3 - shared alpha	4.1	5.7	6.4	7.9	5.7	-
U3 - dose-specific alpha	4.1	5.7	6.5	7.9	5.8	-
U3 - cycle-1 oracle	4.1	6	6.4	7.8	5.8	-
Mean IPDE administrations						
U2 - shared alpha	0.5	2.8	1.6	2	1.5	-
U2 - dose-specific alpha	0.5	2.9	1.6	2	1.5	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.5	2.8	1.7	2	1.5	-
U3 - dose-specific alpha	0.5	2.9	1.6	2	1.5	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1185.1
U2 - dose-specific alpha						1185.1
U2 - cycle-1 oracle						1662.3
U3 - shared alpha						1184.3
U3 - dose-specific alpha						1185.5
U3 - cycle-1 oracle						1662.3
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
MTD selection (%)						
U2 - shared alpha	0.7	0.7	7.6	37.4	53.6	0
U2 - dose-specific alpha	0.7	0.5	7.9	37.5	53.4	0
U2 - cycle-1 oracle	0.7	0.3	7.4	33.8	57.8	0
U3 - shared alpha	0.7	0.5	8.1	36.5	54.2	0
U3 - dose-specific alpha	0.7	0.5	7.7	37.4	53.7	0
U3 - cycle-1 oracle	0.7	0.5	7.9	33.2	57.7	0
OBD selection (%)						
U2 - shared alpha	8.4	12.8	37.1	28.4	12.5	0.8
U2 - dose-specific alpha	8.4	14.5	34.9	28.2	13.2	0.8
U2 - cycle-1 oracle	8.8	14.5	34.4	28.9	12.6	0.8
U3 - shared alpha	13.5	18.5	36.1	25	6.1	0.8
U3 - dose-specific alpha	13.5	19.4	33.7	25.8	6.8	0.8
U3 - cycle-1 oracle	14.1	18.4	33.3	24.5	8.8	0.9
Mean administrations						
U2 - shared alpha	3.9	4.3	7.8	7.3	6.5	-
U2 - dose-specific alpha	4	4.3	7.7	7.3	6.6	-
U2 - cycle-1 oracle	3.9	4.4	7.7	7.4	6.5	-
U3 - shared alpha	4	4.3	7.8	7.3	6.5	-
U3 - dose-specific alpha	4	4.3	7.7	7.3	6.5	-
U3 - cycle-1 oracle	3.9	4.5	7.7	7.3	6.5	-
Mean IPDE administrations						
U2 - shared alpha	0.4	2.8	2.5	2.8	2.2	-
U2 - dose-specific alpha	0.4	2.9	2.5	2.7	2.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.4	2.8	2.5	2.7	2.2	-
U3 - dose-specific alpha	0.4	2.8	2.5	2.7	2.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1056.2
U2 - dose-specific alpha						1056.7
U2 - cycle-1 oracle						1656.6
U3 - shared alpha						1054.6
U3 - dose-specific alpha						1054.9
U3 - cycle-1 oracle						1656.6

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
MTD selection (%)						
U2 - shared alpha	0	0.8	9.6	43.5	46.1	0
U2 - dose-specific alpha	0	0.6	9.1	43.4	46.9	0
U2 - cycle-1 oracle	0	0.7	8.8	41	49.5	0
U3 - shared alpha	0	0.7	9.4	43.6	46.3	0
U3 - dose-specific alpha	0	0.4	10.1	42.7	46.8	0
U3 - cycle-1 oracle	0	0.7	9.4	39.9	50	0
OBD selection (%)						
U2 - shared alpha	7.9	7.2	13.7	41.9	29.2	0.1
U2 - dose-specific alpha	7.9	6.8	13.3	42	29.8	0.2
U2 - cycle-1 oracle	9	4.2	14.3	42.3	30.1	0.1
U3 - shared alpha	12.4	8.9	14	39.8	24.8	0.1
U3 - dose-specific alpha	13.5	9	12.7	39.3	25.4	0.1
U3 - cycle-1 oracle	14.1	6.5	14.3	40.1	24.8	0.2
Mean administrations						
U2 - shared alpha	3.5	3.7	5.3	8.2	9.4	-
U2 - dose-specific alpha	3.5	3.6	5.3	8.1	9.5	-
U2 - cycle-1 oracle	3.5	3.7	5.3	8.6	8.9	-
U3 - shared alpha	3.5	3.6	5.2	8.2	9.4	-
U3 - dose-specific alpha	3.5	3.6	5.3	8.2	9.5	-
U3 - cycle-1 oracle	3.5	3.6	5.4	8.6	8.9	-
Mean IPDE administrations						
U2 - shared alpha	0.2	2.7	1.6	3.1	2.4	-
U2 - dose-specific alpha	0.2	2.6	1.6	3.1	2.4	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.2	2.6	1.6	3.1	2.3	-
U3 - dose-specific alpha	0.2	2.6	1.6	3.1	2.4	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1105.5
U2 - dose-specific alpha						1105.4
U2 - cycle-1 oracle						1663.8
U3 - shared alpha						1106.4
U3 - dose-specific alpha						1106
U3 - cycle-1 oracle						1663.3
Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
MTD selection (%)						
U2 - shared alpha	3.2	11.7	46.8	26.5	11.8	0
U2 - dose-specific alpha	3.3	11.6	46.7	26.5	11.9	0
U2 - cycle-1 oracle	2.6	9.6	43.6	28.4	15.8	0
U3 - shared alpha	3.3	11.4	47.1	26.7	11.5	0
U3 - dose-specific alpha	3.2	10.5	47.2	27.6	11.5	0
U3 - cycle-1 oracle	2.5	10.7	42.3	28.9	15.6	0
OBD selection (%)						
U2 - shared alpha	13.8	12.8	49.5	15.6	3.4	4.9
U2 - dose-specific alpha	13.3	13	48.7	15.6	4.2	5.2
U2 - cycle-1 oracle	13.3	10.5	49.1	17.9	5.1	4.1
U3 - shared alpha	18.6	13.5	46.9	13.5	2.2	5.3
U3 - dose-specific alpha	17.7	14.5	46.3	14.5	2.2	4.8
U3 - cycle-1 oracle	19.6	13.9	44	15.4	2.8	4.3
Mean administrations						
U2 - shared alpha	5.1	5.9	10	5.3	2.8	-
U2 - dose-specific alpha	5.2	5.9	9.9	5.3	2.8	-
U2 - cycle-1 oracle	5	5.6	9.9	5.7	2.9	-
U3 - shared alpha	5.2	5.9	10	5.2	2.8	-
U3 - dose-specific alpha	5.1	5.8	10	5.3	2.9	-
U3 - cycle-1 oracle	5	5.8	10	5.5	3	-
Mean IPDE administrations						
U2 - shared alpha	1	3.3	3.2	1.8	0.9	-
U2 - dose-specific alpha	1	3.3	3.1	1.9	0.9	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1	3.3	3.2	1.8	0.9	-
U3 - dose-specific alpha	1	3.3	3.2	1.9	0.9	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1039.1
U2 - dose-specific alpha						1039.3
U2 - cycle-1 oracle						1618.7
U3 - shared alpha						1039.3
U3 - dose-specific alpha						1040.2
U3 - cycle-1 oracle						1620.9

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
MTD selection (%)						
U2 - shared alpha	1.6	8.3	46.7	31	12.4	0
U2 - dose-specific alpha	1.5	7.7	46.8	30.6	13.4	0
U2 - cycle-1 oracle	1.3	6.7	43.8	31.4	16.8	0
U3 - shared alpha	1.6	8.2	48.2	28.9	13.1	0
U3 - dose-specific alpha	1.6	7.3	48.1	30.2	12.8	0
U3 - cycle-1 oracle	1.4	6.6	44.3	30.9	16.8	0
OBD selection (%)						
U2 - shared alpha	11	42.5	33	10.8	2.5	0.2
U2 - dose-specific alpha	11.8	40.7	34	10.4	2.9	0.2
U2 - cycle-1 oracle	13	41.1	30.3	11.8	3.6	0.2
U3 - shared alpha	15.8	47.4	26.1	8.8	1.7	0.2
U3 - dose-specific alpha	15.6	47.5	27.1	8	1.6	0.2
U3 - cycle-1 oracle	17.1	45.3	25	10.3	2.1	0.2
Mean administrations						
U2 - shared alpha	4.5	8.8	9	5.1	2.7	-
U2 - dose-specific alpha	4.6	8.6	9.1	5	2.7	-
U2 - cycle-1 oracle	4.5	8.6	8.9	5.2	2.7	-
U3 - shared alpha	4.5	8.7	9	5.1	2.6	-
U3 - dose-specific alpha	4.5	8.6	9	5.1	2.7	-
U3 - cycle-1 oracle	4.5	8.8	8.8	5.1	2.8	-
Mean IPDE administrations						
U2 - shared alpha	0.6	3.8	2.2	1.5	0.7	-
U2 - dose-specific alpha	0.6	3.7	2.3	1.5	0.7	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.6	3.8	2.3	1.5	0.7	-
U3 - dose-specific alpha	0.6	3.7	2.2	1.5	0.7	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1165.9
U2 - dose-specific alpha						1165.7
U2 - cycle-1 oracle						1662
U3 - shared alpha						1165.3
U3 - dose-specific alpha						1165
U3 - cycle-1 oracle						1662
Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
MTD selection (%)						
U2 - shared alpha	31.4	34	21.6	10.1	2.5	0.4
U2 - dose-specific alpha	31	35.2	21.4	9.3	2.7	0.4
U2 - cycle-1 oracle	27.7	32.3	23.1	11.8	4.7	0.4
U3 - shared alpha	31.5	34.3	21.3	10.1	2.4	0.4
U3 - dose-specific alpha	31.1	34.3	21.9	9.5	2.8	0.4
U3 - cycle-1 oracle	28.2	31.5	23.9	11.3	4.7	0.4
OBD selection (%)						
U2 - shared alpha	16.5	37.1	12.9	5.3	0.8	27.4
U2 - dose-specific alpha	17.2	37.4	12.9	4.5	1	27
U2 - cycle-1 oracle	16.2	37.1	14.1	5.6	1.6	25.4
U3 - shared alpha	20.6	36.2	10.6	4.3	0.7	27.6
U3 - dose-specific alpha	20.2	37.1	10.7	4.1	0.8	27.1
U3 - cycle-1 oracle	20.7	36.4	11.2	4.9	1.1	25.7
Mean administrations						
U2 - shared alpha	8.3	8.4	5.2	2.3	1	-
U2 - dose-specific alpha	8.2	8.5	5.2	2.3	0.9	-
U2 - cycle-1 oracle	7.7	8.1	5.8	2.8	1.2	-
U3 - shared alpha	8.3	8.4	5.2	2.4	0.9	-
U3 - dose-specific alpha	8.3	8.3	5.3	2.4	1	-
U3 - cycle-1 oracle	7.7	8.2	5.7	2.8	1.2	-
Mean IPDE administrations						
U2 - shared alpha	2.2	4	1.7	0.9	0.3	-
U2 - dose-specific alpha	2.1	4	1.7	0.9	0.3	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.2	4	1.7	0.9	0.3	-
U3 - dose-specific alpha	2.2	3.9	1.7	0.9	0.3	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						890.1
U2 - dose-specific alpha						890.1
U2 - cycle-1 oracle						1424.4
U3 - shared alpha						892.1
U3 - dose-specific alpha						890.9
U3 - cycle-1 oracle						1424.8

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
MTD selection (%)						
U2 - shared alpha	23.5	44.8	24.5	6.3	0.8	0.1
U2 - dose-specific alpha	23.7	43.7	25.5	6.3	0.7	0.1
U2 - cycle-1 oracle	22.5	40.5	27.2	7.5	2.2	0.1
U3 - shared alpha	23.7	45	24.1	6.4	0.7	0.1
U3 - dose-specific alpha	24.4	44	23.7	7.1	0.7	0.1
U3 - cycle-1 oracle	22.1	41.1	26.4	8	2.3	0.1
OBD selection (%)						
U2 - shared alpha	21	47.2	12	1.5	0.2	18.1
U2 - dose-specific alpha	21.1	46.4	12.1	2.1	0.2	18.1
U2 - cycle-1 oracle	21.3	44.4	12.6	3.4	0.5	17.8
U3 - shared alpha	25.4	46	9.4	1.2	0	18
U3 - dose-specific alpha	26.7	45	9	1.7	0	17.6
U3 - cycle-1 oracle	26.5	43	9.9	2.6	0.2	17.8
Mean administrations						
U2 - shared alpha	7.7	10.6	5.8	2	0.7	-
U2 - dose-specific alpha	7.7	10.6	5.9	1.9	0.7	-
U2 - cycle-1 oracle	7.3	10.4	6.2	2.3	0.7	-
U3 - shared alpha	7.7	10.6	5.9	1.9	0.7	-
U3 - dose-specific alpha	7.7	10.7	5.8	1.9	0.7	-
U3 - cycle-1 oracle	7.3	10.3	6.2	2.3	0.8	-
Mean IPDE administrations						
U2 - shared alpha	1.9	4.9	1.8	0.8	0.2	-
U2 - dose-specific alpha	1.9	4.9	1.8	0.8	0.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.9	4.9	1.8	0.8	0.2	-
U3 - dose-specific alpha	1.9	5	1.8	0.8	0.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						940.9
U2 - cycle-1 oracle						940.5
U3 - shared alpha						1495
U3 - dose-specific alpha						939.8
U3 - cycle-1 oracle						940.7
						1491.6

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
MTD selection (%)						
U2 - shared alpha	63.3	18.3	5.8	1.4	0.1	11.1
U2 - dose-specific alpha	63.2	18.7	5.4	1.4	0.1	11.2
U2 - cycle-1 oracle	59	19.9	8	3.9	0.5	8.7
U3 - shared alpha	63.2	18.4	5.5	1.4	0.1	11.4
U3 - dose-specific alpha	62.5	19.5	5.3	1.3	0.1	11.3
U3 - cycle-1 oracle	59.5	19.6	7.8	3.8	0.5	8.8
OBD selection (%)						
U2 - shared alpha	15.2	8	1.6	0.4	0.1	74.7
U2 - dose-specific alpha	16	7.6	1.6	0.2	0.1	74.5
U2 - cycle-1 oracle	15.8	10.3	2.1	0.6	0.1	71.1
U3 - shared alpha	15	9.3	1.2	0.1	0	74.4
U3 - dose-specific alpha	14.9	9.3	1.3	0.1	0	74.4
U3 - cycle-1 oracle	16.1	9.9	1.7	0.5	0	71.8
Mean administrations						
U2 - shared alpha	11.2	5.3	1.9	1	0.2	-
U2 - dose-specific alpha	11.1	5.3	1.9	1	0.2	-
U2 - cycle-1 oracle	10.7	5.6	2.4	1.4	0.4	-
U3 - shared alpha	11.3	5.2	1.9	1	0.2	-
U3 - dose-specific alpha	11.3	5.2	1.9	0.9	0.2	-
U3 - cycle-1 oracle	10.8	5.5	2.3	1.3	0.4	-
Mean IPDE administrations						
U2 - shared alpha	3	2.8	0.6	0.4	0.1	-
U2 - dose-specific alpha	3	2.8	0.6	0.4	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	3.1	2.7	0.6	0.4	0.1	-
U3 - dose-specific alpha	3.1	2.7	0.6	0.4	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						688.8
U2 - dose-specific alpha						688.5
U2 - cycle-1 oracle						1123.2
U3 - shared alpha						686.5
U3 - dose-specific alpha						687.8
U3 - cycle-1 oracle						1120.9
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
MTD selection (%)						
U2 - shared alpha	7.8	31.3	41.6	16.6	2.4	0.3
U2 - dose-specific alpha	7.5	31.7	41	17.1	2.4	0.3
U2 - cycle-1 oracle	5.6	29.2	40.7	19.3	4.9	0.3
U3 - shared alpha	7.6	32.8	41.8	15.6	1.9	0.3
U3 - dose-specific alpha	7.8	33	41.5	15.5	1.9	0.3
U3 - cycle-1 oracle	5.5	31.5	40.4	18.4	3.9	0.3
OBD selection (%)						
U2 - shared alpha	26.9	19.4	38.3	11.5	1	2.9
U2 - dose-specific alpha	27.1	20.1	37.4	11.4	1.1	2.9
U2 - cycle-1 oracle	25.8	18.5	38.5	12.2	2.1	2.9
U3 - shared alpha	31.1	19.8	35.2	10.2	0.9	2.8
U3 - dose-specific alpha	30.9	20.6	34.7	10	0.9	2.9
U3 - cycle-1 oracle	29.3	19.7	35.5	11.6	1.2	2.7
Mean administrations						
U2 - shared alpha	9.6	7.6	8.8	3.1	0.5	-
U2 - dose-specific alpha	9.6	7.5	8.8	3.1	0.5	-
U2 - cycle-1 oracle	9.1	7	9	3.7	0.8	-
U3 - shared alpha	10.1	7.7	8.5	2.8	0.4	-
U3 - dose-specific alpha	10.3	7.7	8.4	2.8	0.4	-
U3 - cycle-1 oracle	9.7	7.2	8.6	3.4	0.7	-
Mean IPDE administrations						
U2 - shared alpha	2.4	3.4	2.4	0.9	0.1	-
U2 - dose-specific alpha	2.4	3.4	2.4	0.9	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.6	3.5	2.4	0.9	0.1	-
U3 - dose-specific alpha	2.7	3.5	2.3	0.9	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1113
U2 - dose-specific alpha						1115.2
U2 - cycle-1 oracle						1639.8
U3 - shared alpha						1111.1
U3 - dose-specific alpha						1110.9
U3 - cycle-1 oracle						1642.2

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
MTD selection (%)						
U2 - shared alpha	11.3	31.1	36.1	17.8	3.4	0.3
U2 - dose-specific alpha	10.9	31.8	36.4	16.8	3.8	0.3
U2 - cycle-1 oracle	9.5	26.9	35	21.3	7	0.3
U3 - shared alpha	11.1	35.6	34.6	15.3	3.1	0.3
U3 - dose-specific alpha	11.3	35	35.4	15.2	2.8	0.3
U3 - cycle-1 oracle	9.7	31	34.2	18.9	5.9	0.3
OBD selection (%)						
U2 - shared alpha	8	47	24.8	9.7	1.3	9.2
U2 - dose-specific alpha	8.2	47.6	24.3	9.6	1.4	8.9
U2 - cycle-1 oracle	7.8	46.1	23.8	11.6	2.6	8.1
U3 - shared alpha	9.2	51.6	21.9	7.2	1	9.1
U3 - dose-specific alpha	9.3	53.2	20.5	6.8	1	9.2
U3 - cycle-1 oracle	9.8	50.5	21	8.8	1.5	8.4
Mean administrations						
U2 - shared alpha	6.8	11.6	6.6	3	0.7	-
U2 - dose-specific alpha	6.8	11.7	6.5	3	0.7	-
U2 - cycle-1 oracle	6.2	11.3	6.7	3.6	1	-
U3 - shared alpha	6.9	12.4	6.1	2.7	0.5	-
U3 - dose-specific alpha	6.9	12.5	6.1	2.7	0.5	-
U3 - cycle-1 oracle	6.4	12.1	6.3	3.2	0.8	-
Mean IPDE administrations						
U2 - shared alpha	1.3	4.7	2	1	0.2	-
U2 - dose-specific alpha	1.3	4.8	1.9	1	0.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.4	5	1.8	0.9	0.2	-
U3 - dose-specific alpha	1.3	5.1	1.8	0.9	0.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1070
U2 - dose-specific alpha						1069.8
U2 - cycle-1 oracle						1599.2
U3 - shared alpha						1065.9
U3 - dose-specific alpha						1066.2
U3 - cycle-1 oracle						1598.5
Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
MTD selection (%)						
U2 - shared alpha	0.8	2.8	18.9	58.6	18.9	0
U2 - dose-specific alpha	0.8	2.5	18.7	59.7	18.3	0
U2 - cycle-1 oracle	0.7	2.2	16.8	54.5	25.8	0
U3 - shared alpha	0.8	2.8	19.4	61	16	0
U3 - dose-specific alpha	0.8	2.7	19.3	60.6	16.6	0
U3 - cycle-1 oracle	0.7	2.8	16.5	57.3	22.7	0
OBD selection (%)						
U2 - shared alpha	6.7	5.4	18.8	55.1	12.4	1.6
U2 - dose-specific alpha	6.3	5.7	19.3	54.2	12.8	1.7
U2 - cycle-1 oracle	6.3	5.1	17.3	53.9	15.8	1.6
U3 - shared alpha	8.6	6.8	19.4	54	9.6	1.6
U3 - dose-specific alpha	8.4	7.5	18.9	53.5	10.1	1.6
U3 - cycle-1 oracle	9.3	7.9	16.2	52.9	12.1	1.6
Mean administrations						
U2 - shared alpha	4.2	4.7	7.1	10.7	3.1	-
U2 - dose-specific alpha	4.2	4.7	7.1	10.8	3	-
U2 - cycle-1 oracle	4.1	4.6	6.8	10.7	3.6	-
U3 - shared alpha	4.4	4.8	7.2	10.8	2.7	-
U3 - dose-specific alpha	4.3	4.8	7.2	10.8	2.7	-
U3 - cycle-1 oracle	4.3	4.7	6.8	10.8	3.2	-
Mean IPDE administrations						
U2 - shared alpha	0.5	3.1	2.1	3.4	0.9	-
U2 - dose-specific alpha	0.5	3.1	2.1	3.4	0.8	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.6	3.1	2.1	3.4	0.7	-
U3 - dose-specific alpha	0.6	3.1	2.1	3.4	0.7	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1098.6
U2 - dose-specific alpha						1098.5
U2 - cycle-1 oracle						1653.8
U3 - shared alpha						1093.2
U3 - dose-specific alpha						1093.9
U3 - cycle-1 oracle						1652.4

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
MTD selection (%)						
U2 - shared alpha	1.5	3.7	16.9	33.5	44.4	0
U2 - dose-specific alpha	1.5	3.6	17.7	33	44.2	0
U2 - cycle-1 oracle	1.5	3.5	14.5	28.4	52.1	0
U3 - shared alpha	1.5	6	19.2	34.6	38.7	0
U3 - dose-specific alpha	1.5	5.4	18.8	33.7	40.6	0
U3 - cycle-1 oracle	1.5	5.1	17.5	30.9	45	0
OBD selection (%)						
U2 - shared alpha	5.2	33.4	26.4	23.1	10.4	1.5
U2 - dose-specific alpha	5.4	33.8	26.1	22.9	10.3	1.5
U2 - cycle-1 oracle	5.5	39.1	24.2	20.2	9.5	1.5
U3 - shared alpha	6.9	44.3	23.2	18.4	5.7	1.5
U3 - dose-specific alpha	7.5	42.1	23.3	19	6.6	1.5
U3 - cycle-1 oracle	9.1	43.9	23.4	16.6	5.5	1.5
Mean administrations						
U2 - shared alpha	4	7.5	7.2	7.2	3.9	-
U2 - dose-specific alpha	4	7.5	7.2	7.1	3.9	-
U2 - cycle-1 oracle	3.8	7.4	7.1	7.1	4.4	-
U3 - shared alpha	4.1	8.2	7.4	6.8	3.3	-
U3 - dose-specific alpha	4.1	8.2	7.3	6.9	3.3	-
U3 - cycle-1 oracle	3.9	8.1	7.2	6.8	3.8	-
Mean IPDE administrations						
U2 - shared alpha	0.3	3.9	2.3	2.7	1.4	-
U2 - dose-specific alpha	0.3	3.9	2.3	2.7	1.4	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.4	4.2	2.4	2.6	1.2	-
U3 - dose-specific alpha	0.4	4.2	2.4	2.6	1.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - shared alpha						1047.9
U2 - dose-specific alpha						1047.7
U2 - cycle-1 oracle						1650
U3 - shared alpha						1046.7
U3 - dose-specific alpha						1046.3
U3 - cycle-1 oracle						1650
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
MTD selection (%)						
U2 - shared alpha	53.1	23	8.3	2.4	0.2	13
U2 - dose-specific alpha	52.8	23.5	8.7	2.1	0.2	12.7
U2 - cycle-1 oracle	45.3	27.6	10.5	4.2	0.5	11.9
U3 - shared alpha	53.8	23.5	7.3	2.1	0.2	13.1
U3 - dose-specific alpha	53.1	23.9	7.4	2.1	0.2	13.3
U3 - cycle-1 oracle	45.2	29.4	9	3.9	0.6	11.9
OBD selection (%)						
U2 - shared alpha	66	12.8	3.6	1.1	0.2	16.3
U2 - dose-specific alpha	65.2	13.5	3.6	1.1	0.2	16.4
U2 - cycle-1 oracle	60.7	17	4.6	2	0.4	15.3
U3 - shared alpha	66.4	12.4	3	1.1	0.2	16.9
U3 - dose-specific alpha	67.4	12.5	3	0.9	0.2	16
U3 - cycle-1 oracle	62.5	16.6	3.5	1.6	0.3	15.5
Mean administrations						
U2 - shared alpha	18.3	5.7	2.2	0.8	0.2	-
U2 - dose-specific alpha	18.3	5.7	2.2	0.8	0.2	-
U2 - cycle-1 oracle	17.2	6.1	2.7	1.3	0.3	-
U3 - shared alpha	18.7	5.6	1.9	0.7	0.2	-
U3 - dose-specific alpha	18.9	5.6	2	0.7	0.2	-
U3 - cycle-1 oracle	17.7	5.9	2.4	1.1	0.2	-
Mean IPDE administrations						
U2 - shared alpha	4.7	2.3	0.7	0.3	0.1	-
U2 - dose-specific alpha	4.7	2.3	0.7	0.3	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	4.9	2.3	0.6	0.3	0.1	-
U3 - dose-specific alpha	4.9	2.3	0.6	0.3	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - shared alpha						1059
U2 - dose-specific alpha						1058.4
U2 - cycle-1 oracle						1524.7
U3 - shared alpha						1054.9
U3 - dose-specific alpha						1059.8
U3 - cycle-1 oracle						1522.1

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
MTD selection (%)						
U2 - shared alpha	8.8	46	29.9	11.7	3.1	0.5
U2 - dose-specific alpha	8.6	46.5	29.1	12.2	3.2	0.4
U2 - cycle-1 oracle	6.9	41.3	29.3	15.6	6.5	0.4
U3 - shared alpha	8.7	53.1	24.7	10.5	2.6	0.4
U3 - dose-specific alpha	8.7	52.6	25.5	10.3	2.5	0.4
U3 - cycle-1 oracle	6.8	47.3	26.6	14.1	4.8	0.4
OBD selection (%)						
U2 - shared alpha	12.4	51	24.4	7.8	1.5	2.9
U2 - dose-specific alpha	12.2	51.6	23.5	8	1.8	2.9
U2 - cycle-1 oracle	11.7	48.5	23.6	10.1	3.3	2.8
U3 - shared alpha	14.2	55.8	19	6.8	1.2	3
U3 - dose-specific alpha	14	55.1	20.1	6.8	1.2	2.8
U3 - cycle-1 oracle	14.1	53.2	19.7	8.4	1.8	2.8
Mean administrations						
U2 - shared alpha	7.9	13.7	5.6	2	0.4	-
U2 - dose-specific alpha	7.9	13.5	5.7	2	0.4	-
U2 - cycle-1 oracle	7.5	12.8	5.8	2.7	0.8	-
U3 - shared alpha	8.2	14.2	5	1.8	0.3	-
U3 - dose-specific alpha	8.3	14.2	5	1.8	0.3	-
U3 - cycle-1 oracle	7.9	13.5	5.2	2.4	0.6	-
Mean IPDE administrations						
U2 - shared alpha	1.6	4.9	1.6	0.6	0.1	-
U2 - dose-specific alpha	1.6	4.9	1.6	0.6	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.7	5.1	1.3	0.6	0.1	-
U3 - dose-specific alpha	1.8	5.1	1.3	0.6	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1145.7
U2 - dose-specific alpha						1145.8
U2 - cycle-1 oracle						1639.8
U3 - shared alpha						1143.1
U3 - dose-specific alpha						1145
U3 - cycle-1 oracle						1639.8

Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
MTD selection (%)						
U2 - shared alpha	62.9	13.9	4	1.1	0.3	17.8
U2 - dose-specific alpha	63.7	13.5	4.4	0.9	0.3	17.2
U2 - cycle-1 oracle	58.9	17.6	7	1.9	0.5	14.1
U3 - shared alpha	63	13.8	4.1	0.9	0.3	17.9
U3 - dose-specific alpha	63.7	14.1	3.8	0.8	0.3	17.3
U3 - cycle-1 oracle	59.5	17.6	6.6	1.8	0.5	14
OBD selection (%)						
U2 - shared alpha	29.4	8.3	1.7	0.4	0.2	60
U2 - dose-specific alpha	28.3	8.9	1.6	0.3	0.2	60.7
U2 - cycle-1 oracle	27.4	11.7	3	1	0.2	56.7
U3 - shared alpha	29.4	8.1	1.6	0.3	0.2	60.4
U3 - dose-specific alpha	29.2	8.5	1.3	0.3	0.2	60.5
U3 - cycle-1 oracle	28.2	11.5	2.5	0.8	0.2	56.8
Mean administrations						
U2 - shared alpha	13.4	5.3	1.8	0.8	0.1	-
U2 - dose-specific alpha	13.3	5.4	1.8	0.8	0.2	-
U2 - cycle-1 oracle	12.7	5.8	2.3	1.1	0.2	-
U3 - shared alpha	13.6	5.2	1.6	0.7	0.1	-
U3 - dose-specific alpha	13.6	5.2	1.6	0.7	0.1	-
U3 - cycle-1 oracle	12.8	5.7	2.2	1.1	0.2	-
Mean IPDE administrations						
U2 - shared alpha	3.6	2.6	0.6	0.3	0.1	-
U2 - dose-specific alpha	3.6	2.6	0.6	0.3	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	3.7	2.5	0.5	0.3	0	-
U3 - dose-specific alpha	3.7	2.6	0.5	0.3	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						775.8
U2 - dose-specific alpha						774.6
U2 - cycle-1 oracle						1221.1
U3 - shared alpha						773.1
U3 - dose-specific alpha						774.9
U3 - cycle-1 oracle						1214.6

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
MTD selection (%)						
U2 - shared alpha	11.7	12.6	19.3	26.7	29.4	0.3
U2 - dose-specific alpha	11.4	13.5	18.5	26.9	29.4	0.3
U2 - cycle-1 oracle	9.3	11	16.9	25.7	36.8	0.3
U3 - shared alpha	12.1	13.4	18.3	27.5	28.4	0.3
U3 - dose-specific alpha	11.5	14.3	18	28	27.9	0.3
U3 - cycle-1 oracle	9.3	11.5	17.3	25.9	35.7	0.3
OBD selection (%)						
U2 - shared alpha	11.6	11.7	21.5	17.5	21.8	15.9
U2 - dose-specific alpha	12.3	12.1	20.8	16.4	22.1	16.3
U2 - cycle-1 oracle	11	11.1	21.2	16.6	26.1	14
U3 - shared alpha	15.5	13.9	19.7	16	17.8	17.1
U3 - dose-specific alpha	15.9	14.5	19.9	15.2	17.8	16.7
U3 - cycle-1 oracle	15.9	13.9	19.4	14	22.6	14.2
Mean administrations						
U2 - shared alpha	6.7	5.8	6	5.2	4.2	-
U2 - dose-specific alpha	6.7	5.9	6	5.1	4.2	-
U2 - cycle-1 oracle	6.2	5.7	6	5.5	4.8	-
U3 - shared alpha	6.9	5.9	6	5.2	3.9	-
U3 - dose-specific alpha	6.9	6	6	5.1	3.9	-
U3 - cycle-1 oracle	6.4	5.8	6	5.4	4.7	-
Mean IPDE administrations						
U2 - shared alpha	1.5	3.4	2	2.1	1.3	-
U2 - dose-specific alpha	1.5	3.4	2	2	1.3	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.6	3.4	2	2	1.2	-
U3 - dose-specific alpha	1.6	3.4	2	2	1.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						977.8
U2 - cycle-1 oracle						978
U3 - shared alpha						1570.4
U3 - dose-specific alpha						974.9
U3 - cycle-1 oracle						974.7
						1566.3
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
MTD selection (%)						
U2 - shared alpha	44.2	25.9	18.6	7.3	1.3	2.7
U2 - dose-specific alpha	44.2	25.6	19.3	6.9	1.2	2.8
U2 - cycle-1 oracle	37.3	25	21	11.6	2.7	2.4
U3 - shared alpha	44	27.7	18	6.6	1.1	2.6
U3 - dose-specific alpha	43.8	28.2	17.8	6.5	1.1	2.6
U3 - cycle-1 oracle	37.6	26.1	21.2	10.3	2.3	2.5
OBD selection (%)						
U2 - shared alpha	7.6	31.5	14.2	4	0.9	41.8
U2 - dose-specific alpha	7.6	32.5	13.5	3.6	0.9	41.9
U2 - cycle-1 oracle	5.2	34.9	15.3	6.7	1.3	36.6
U3 - shared alpha	8.1	33.7	12.5	3.4	0.8	41.5
U3 - dose-specific alpha	7.9	34	12.7	3.2	0.8	41.4
U3 - cycle-1 oracle	5.9	37.5	13.9	5.2	0.9	36.6
Mean administrations						
U2 - shared alpha	8.6	9.6	4	1.6	0.4	-
U2 - dose-specific alpha	8.6	9.6	4	1.6	0.4	-
U2 - cycle-1 oracle	7.7	9.8	4.7	2.1	0.6	-
U3 - shared alpha	8.7	9.9	3.8	1.5	0.3	-
U3 - dose-specific alpha	8.7	9.9	3.9	1.4	0.3	-
U3 - cycle-1 oracle	7.8	10.2	4.4	2	0.5	-
Mean IPDE administrations						
U2 - shared alpha	1.8	4.1	1.1	0.5	0.1	-
U2 - dose-specific alpha	1.8	4.1	1.1	0.6	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.9	4.2	1.1	0.5	0.1	-
U3 - dose-specific alpha	1.9	4.2	1.1	0.5	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						903.3
U2 - cycle-1 oracle						902
U3 - shared alpha						1377.8
U3 - dose-specific alpha						900.6
U3 - cycle-1 oracle						901.6
						1374.4

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
MTD selection (%)						
U2 - shared alpha	41.3	43.1	11.6	0.7	0	3.3
U2 - dose-specific alpha	40.5	43.3	11.9	1	0	3.3
U2 - cycle-1 oracle	36.5	43.8	14.6	2.4	0	2.7
U3 - shared alpha	41.7	43.4	10.7	0.8	0	3.4
U3 - dose-specific alpha	40.9	43.9	11.4	0.5	0	3.3
U3 - cycle-1 oracle	35.1	45.1	14.9	2	0	2.9
OBD selection (%)						
U2 - shared alpha	60.9	21.5	3.3	0.1	0	14.2
U2 - dose-specific alpha	60.4	20.9	4.2	0.2	0	14.3
U2 - cycle-1 oracle	59.3	21.3	4.1	0.4	0	14.9
U3 - shared alpha	62.6	20.6	2.4	0.1	0	14.3
U3 - dose-specific alpha	62.6	20.4	2.7	0.1	0	14.2
U3 - cycle-1 oracle	60.5	19.8	3.8	0.3	0	15.6
Mean administrations						
U2 - shared alpha	16.8	7.5	2.7	1	0.1	-
U2 - dose-specific alpha	16.8	7.4	2.7	1	0.1	-
U2 - cycle-1 oracle	15.7	7.7	3.3	1.3	0.2	-
U3 - shared alpha	17.2	7.4	2.5	0.9	0.1	-
U3 - dose-specific alpha	17.2	7.4	2.6	0.9	0.1	-
U3 - cycle-1 oracle	16.2	7.4	3	1.2	0.2	-
Mean IPDE administrations						
U2 - shared alpha	4.9	3.4	0.8	0.4	0	-
U2 - dose-specific alpha	4.9	3.4	0.9	0.4	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	5	3.4	0.8	0.3	0	-
U3 - dose-specific alpha	5	3.4	0.8	0.3	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1022.3
U2 - dose-specific alpha						1021.9
U2 - cycle-1 oracle						1564.7
U3 - shared alpha						1020.2
U3 - dose-specific alpha						1021.8
U3 - cycle-1 oracle						1562.1
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
MTD selection (%)						
U2 - shared alpha	1.1	2.6	17.8	56.6	21.9	0
U2 - dose-specific alpha	1.2	2.7	17.7	55.9	22.5	0
U2 - cycle-1 oracle	0.9	2.5	16.4	49.8	30.4	0
U3 - shared alpha	1.1	2.7	18	56.8	21.4	0
U3 - dose-specific alpha	1.1	2.7	18.2	57.7	20.3	0
U3 - cycle-1 oracle	0.9	2.5	16.8	51.3	28.5	0
OBD selection (%)						
U2 - shared alpha	11.4	10.8	11.9	47	16.4	2.5
U2 - dose-specific alpha	11.4	12.4	10.9	46.1	16.7	2.5
U2 - cycle-1 oracle	11.8	11.5	7.7	44.6	22.3	2.1
U3 - shared alpha	15.3	13.8	9.2	43.7	15.5	2.5
U3 - dose-specific alpha	15.6	14.8	9.5	43.7	13.9	2.5
U3 - cycle-1 oracle	14.6	14.8	8.3	40.6	19.5	2.2
Mean administrations						
U2 - shared alpha	4.6	5.3	5.9	9.5	4.5	-
U2 - dose-specific alpha	4.6	5.3	5.9	9.5	4.5	-
U2 - cycle-1 oracle	4.4	5.2	5.7	9.5	5.1	-
U3 - shared alpha	4.8	5.4	5.8	9.5	4.2	-
U3 - dose-specific alpha	4.9	5.5	5.8	9.4	4.2	-
U3 - cycle-1 oracle	4.6	5.3	5.5	9.6	4.8	-
Mean IPDE administrations						
U2 - shared alpha	0.7	3.3	1.7	3.4	1.2	-
U2 - dose-specific alpha	0.7	3.3	1.7	3.3	1.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.8	3.4	1.7	3.4	1.1	-
U3 - dose-specific alpha	0.8	3.4	1.7	3.3	1.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1071.1
U2 - dose-specific alpha						1071.4
U2 - cycle-1 oracle						1651.5
U3 - shared alpha						1068.5
U3 - dose-specific alpha						1067.3
U3 - cycle-1 oracle						1651.3

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
MTD selection (%)						
U2 - shared alpha	40.9	25.1	18.7	11.1	2.2	2
U2 - dose-specific alpha	40.8	25.1	19.1	11	2	2
U2 - cycle-1 oracle	34.8	24.8	19.7	15.6	3.4	1.7
U3 - shared alpha	41	25.9	19.2	10.4	1.5	2
U3 - dose-specific alpha	41.4	25.6	19.1	10.2	1.8	1.9
U3 - cycle-1 oracle	35.2	24.8	20	15.4	2.8	1.8
OBD selection (%)						
U2 - shared alpha	9	32	14.2	4.8	1.2	38.8
U2 - dose-specific alpha	9.4	32.5	14.2	4.1	1.2	38.6
U2 - cycle-1 oracle	7.2	35.8	15	6.3	1.6	34.1
U3 - shared alpha	10.5	32.6	13.8	3.5	0.8	38.8
U3 - dose-specific alpha	10.5	32.9	12.4	4.1	0.9	39.2
U3 - cycle-1 oracle	8.7	36.2	14.9	4.9	1.3	34
Mean administrations						
U2 - shared alpha	8.6	8.8	4.5	2.3	0.6	-
U2 - dose-specific alpha	8.5	8.8	4.5	2.2	0.5	-
U2 - cycle-1 oracle	7.8	9.1	4.8	2.8	0.9	-
U3 - shared alpha	8.7	9.1	4.3	2.1	0.5	-
U3 - dose-specific alpha	8.7	9.1	4.3	2.1	0.5	-
U3 - cycle-1 oracle	7.9	9.4	4.7	2.6	0.7	-
Mean IPDE administrations						
U2 - shared alpha	1.9	4.1	1.4	0.8	0.2	-
U2 - dose-specific alpha	1.9	4.1	1.4	0.8	0.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2	4.2	1.3	0.8	0.2	-
U3 - dose-specific alpha	2	4.2	1.3	0.8	0.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						894.6
U2 - dose-specific alpha						893
U2 - cycle-1 oracle						1402.7
U3 - shared alpha						890.5
U3 - dose-specific alpha						891.4
U3 - cycle-1 oracle						1401.5

Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
MTD selection (%)						
U2 - shared alpha	2.8	10.3	25	38.9	23	0
U2 - dose-specific alpha	2.8	9.8	25.7	38.5	23.2	0
U2 - cycle-1 oracle	2.7	8.4	22.2	36.7	30	0
U3 - shared alpha	2.9	10.1	25.5	39.9	21.6	0
U3 - dose-specific alpha	2.7	9.5	25.5	39.8	22.5	0
U3 - cycle-1 oracle	2.5	8.8	22.2	38.2	28.3	0
OBD selection (%)						
U2 - shared alpha	16.3	17.2	15.7	31.7	12.3	6.8
U2 - dose-specific alpha	15.6	16.7	16.9	31.4	12.6	6.8
U2 - cycle-1 oracle	17.2	15	14.6	32.2	14.7	6.3
U3 - shared alpha	23	20.1	14.6	27	8.3	7
U3 - dose-specific alpha	21.5	19.9	15.5	27.7	8.6	6.8
U3 - cycle-1 oracle	21.7	18.4	13.1	29.7	10.6	6.5
Mean administrations						
U2 - shared alpha	6	6.4	5.9	7.4	3.6	-
U2 - dose-specific alpha	6	6.4	6	7.3	3.7	-
U2 - cycle-1 oracle	5.7	6.1	5.8	7.7	4.1	-
U3 - shared alpha	6.4	6.5	5.8	7.3	3.4	-
U3 - dose-specific alpha	6.4	6.4	5.8	7.3	3.4	-
U3 - cycle-1 oracle	6	6.1	5.7	7.8	3.8	-
Mean IPDE administrations						
U2 - shared alpha	1.3	3.7	1.9	2.8	1.2	-
U2 - dose-specific alpha	1.3	3.7	1.9	2.8	1.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.4	3.8	1.8	2.8	1.1	-
U3 - dose-specific alpha	1.5	3.8	1.9	2.8	1.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1014.7
U2 - dose-specific alpha						1014.6
U2 - cycle-1 oracle						1631.3
U3 - shared alpha						1014.2
U3 - dose-specific alpha						1014.1
U3 - cycle-1 oracle						1630.1

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
MTD selection (%)						
U2 - shared alpha	9	29.6	36.3	19.8	5.1	0.2
U2 - dose-specific alpha	8.9	29.2	37.1	20.3	4.3	0.2
U2 - cycle-1 oracle	7.7	26.9	33.6	23.7	7.9	0.2
U3 - shared alpha	8.9	30.3	35.7	20.3	4.6	0.2
U3 - dose-specific alpha	9	30	35.5	20.4	4.9	0.2
U3 - cycle-1 oracle	8.4	26.5	33.3	23.2	8.4	0.2
OBD selection (%)						
U2 - shared alpha	41.2	29	11.8	4.9	0.5	12.6
U2 - dose-specific alpha	40	29.4	11.9	5.2	0.4	13.1
U2 - cycle-1 oracle	41.8	27.8	11.5	6.2	0.4	12.3
U3 - shared alpha	45.7	30.6	7.3	3.5	0.1	12.8
U3 - dose-specific alpha	46.1	28.6	8.2	3.7	0.3	13.1
U3 - cycle-1 oracle	45.1	29.1	9.3	3.2	0.3	13
Mean administrations						
U2 - shared alpha	9.7	8.5	5.8	3.6	1.2	-
U2 - dose-specific alpha	9.7	8.4	5.8	3.6	1.1	-
U2 - cycle-1 oracle	9.4	8.2	5.7	4	1.5	-
U3 - shared alpha	10.4	8.3	5.5	3.5	1.1	-
U3 - dose-specific alpha	10.4	8.3	5.4	3.5	1.1	-
U3 - cycle-1 oracle	10	8	5.5	3.8	1.4	-
Mean IPDE administrations						
U2 - shared alpha	2.8	4.3	2	1.5	0.4	-
U2 - dose-specific alpha	2.8	4.3	2	1.5	0.4	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	3.1	4.2	1.8	1.5	0.4	-
U3 - dose-specific alpha	3.1	4.3	1.8	1.5	0.4	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						972.4
U2 - dose-specific alpha						971.4
U2 - cycle-1 oracle						1596
U3 - shared alpha						971
U3 - dose-specific alpha						970.4
U3 - cycle-1 oracle						1594.5
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
MTD selection (%)						
U2 - shared alpha	0	1	1.9	11.7	85.4	0
U2 - dose-specific alpha	0	1.6	2	10.9	85.5	0
U2 - cycle-1 oracle	0	1.4	1.1	10.1	87.4	0
U3 - shared alpha	0	2.1	3.8	12.1	82	0
U3 - dose-specific alpha	0	2.5	2.4	13	82.1	0
U3 - cycle-1 oracle	0	1.5	2.1	10.8	85.6	0
OBD selection (%)						
U2 - shared alpha	37.8	27.2	16	12.1	6.9	0
U2 - dose-specific alpha	38.1	26.6	15.2	12.4	7.7	0
U2 - cycle-1 oracle	39.6	24.9	15.7	13.1	6.7	0
U3 - shared alpha	42	30	13.7	10.2	4.1	0
U3 - dose-specific alpha	43.9	27.9	14	10.4	3.8	0
U3 - cycle-1 oracle	45.6	24.1	14.6	11.3	4.4	0
Mean administrations						
U2 - shared alpha	4.3	5	5.9	8.4	6.3	-
U2 - dose-specific alpha	4.4	5.1	5.9	8.3	6.2	-
U2 - cycle-1 oracle	4.5	4.9	6	8.4	6.1	-
U3 - shared alpha	4.7	5.4	6.2	8.1	5.7	-
U3 - dose-specific alpha	4.7	5.3	6.2	8.2	5.6	-
U3 - cycle-1 oracle	4.6	5.2	6.2	8.3	5.7	-
Mean IPDE administrations						
U2 - shared alpha	0.5	2.9	2.3	3.6	2.4	-
U2 - dose-specific alpha	0.6	2.9	2.3	3.6	2.3	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.7	3.1	2.3	3.5	2.1	-
U3 - dose-specific alpha	0.7	3	2.3	3.5	2.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1009.1
U2 - dose-specific alpha						1008.9
U2 - cycle-1 oracle						1664.1
U3 - shared alpha						1007.1
U3 - dose-specific alpha						1006.9
U3 - cycle-1 oracle						1664.1

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
MTD selection (%)						
U2 - shared alpha	5.3	66.4	25.9	2.3	0	0.1
U2 - dose-specific alpha	6.2	65.3	26.5	1.9	0	0.1
U2 - cycle-1 oracle	5.6	59.7	30.3	4	0.3	0.1
U3 - shared alpha	6	70.3	22	1.6	0	0.1
U3 - dose-specific alpha	6	70.3	22	1.6	0	0.1
U3 - cycle-1 oracle	5.2	64.2	27.6	2.9	0	0.1
OBD selection (%)						
U2 - shared alpha	18.4	65.1	14.8	1.2	0	0.5
U2 - dose-specific alpha	20.5	63.5	14.4	1.1	0	0.5
U2 - cycle-1 oracle	19.5	61.2	16.7	1.9	0.1	0.6
U3 - shared alpha	20.9	65.7	12.1	0.6	0	0.7
U3 - dose-specific alpha	22.3	65	11.4	0.7	0	0.6
U3 - cycle-1 oracle	22.8	62.2	13	1.4	0	0.6
Mean administrations						
U2 - shared alpha	9.4	15.4	4.4	0.7	0.1	-
U2 - dose-specific alpha	9.6	15.3	4.3	0.7	0.1	-
U2 - cycle-1 oracle	9.1	14.3	5.2	1.2	0.1	-
U3 - shared alpha	10	15.6	3.8	0.5	0	-
U3 - dose-specific alpha	10.2	15.5	3.6	0.5	0	-
U3 - cycle-1 oracle	9.6	14.7	4.5	1	0.1	-
Mean IPDE administrations						
U2 - shared alpha	2.1	5.1	1.2	0.2	0	-
U2 - dose-specific alpha	2.1	5	1.2	0.2	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.3	5.2	1	0.2	0	-
U3 - dose-specific alpha	2.4	5.2	1	0.2	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1180.1
U2 - dose-specific alpha						1178.3
U2 - cycle-1 oracle						1658.1
U3 - shared alpha						1173.4
U3 - dose-specific alpha						1171.7
U3 - cycle-1 oracle						1658.1
Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
MTD selection (%)						
U2 - shared alpha	0.1	0.1	2.6	25.4	71.8	0
U2 - dose-specific alpha	0.1	0.1	2.3	25.6	71.9	0
U2 - cycle-1 oracle	0	0.1	1.7	23.2	75	0
U3 - shared alpha	0.1	0.1	2.4	25.2	72.2	0
U3 - dose-specific alpha	0.1	0.1	2.7	26	71.1	0
U3 - cycle-1 oracle	0	0.2	1.7	22.7	75.4	0
OBD selection (%)						
U2 - shared alpha	10.3	10.2	8.8	15.7	54.9	0.1
U2 - dose-specific alpha	9.9	10.8	8.9	16.4	53.9	0.1
U2 - cycle-1 oracle	11.3	10.2	9	15.1	54.4	0
U3 - shared alpha	16.2	13.1	10.2	13.3	47.1	0.1
U3 - dose-specific alpha	16.8	14.5	9.8	13.3	45.5	0.1
U3 - cycle-1 oracle	18.4	14	9.2	11.2	47.2	0
Mean administrations						
U2 - shared alpha	3.4	4.1	5.2	7.1	10.2	-
U2 - dose-specific alpha	3.5	4	5.2	7.1	10.2	-
U2 - cycle-1 oracle	3.4	4.1	5.3	7	10.3	-
U3 - shared alpha	3.5	4.2	5.4	7.2	9.7	-
U3 - dose-specific alpha	3.5	4.2	5.3	7.2	9.7	-
U3 - cycle-1 oracle	3.5	4.2	5.3	7	10	-
Mean IPDE administrations						
U2 - shared alpha	0.2	2.8	1.8	3	2.9	-
U2 - dose-specific alpha	0.2	2.8	1.8	2.9	2.9	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.2	2.9	1.9	3	2.8	-
U3 - dose-specific alpha	0.2	2.9	1.8	3	2.8	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1060.7
U2 - dose-specific alpha						1061.8
U2 - cycle-1 oracle						1664.1
U3 - shared alpha						1057.3
U3 - dose-specific alpha						1058.1
U3 - cycle-1 oracle						1664.1

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
MTD selection (%)						
U2 - shared alpha	18.1	47.3	30.4	4.2	0	0
U2 - dose-specific alpha	18.4	46.4	31.4	3.8	0	0
U2 - cycle-1 oracle	18	42.7	33.2	6.1	0	0
U3 - shared alpha	18	47.2	30.7	4.1	0	0
U3 - dose-specific alpha	17.9	47.2	30.9	4	0	0
U3 - cycle-1 oracle	16.8	44.3	32.6	6.2	0.1	0
OBD selection (%)						
U2 - shared alpha	11.7	56.2	16.7	1.2	0	14.2
U2 - dose-specific alpha	11.8	57.2	15.8	0.7	0	14.5
U2 - cycle-1 oracle	12.1	53.8	17.4	2.1	0	14.6
U3 - shared alpha	13.2	57.8	13.6	0.8	0	14.6
U3 - dose-specific alpha	13.7	56.9	14.1	0.6	0	14.7
U3 - cycle-1 oracle	12.8	55.1	16	1.8	0.1	14.2
Mean administrations						
U2 - shared alpha	7.9	12.6	5.9	1.9	0.2	-
U2 - dose-specific alpha	7.9	12.6	5.9	1.9	0.2	-
U2 - cycle-1 oracle	7.4	12.2	6.4	2.2	0.4	-
U3 - shared alpha	8.2	12.8	5.6	1.7	0.2	-
U3 - dose-specific alpha	8.1	12.8	5.7	1.7	0.2	-
U3 - cycle-1 oracle	7.5	12.6	6	2	0.3	-
Mean IPDE administrations						
U2 - shared alpha	1.7	5.5	1.7	0.7	0.1	-
U2 - dose-specific alpha	1.7	5.5	1.7	0.7	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.8	5.6	1.6	0.6	0.1	-
U3 - dose-specific alpha	1.8	5.6	1.7	0.6	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1033.5
U2 - dose-specific alpha						1032.7
U2 - cycle-1 oracle						1578.7
U3 - shared alpha						1029.4
U3 - dose-specific alpha						1029.1
U3 - cycle-1 oracle						1577.3

Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
MTD selection (%)						
U2 - shared alpha	2.2	14.8	49.3	28.6	5.1	0
U2 - dose-specific alpha	2.3	14.4	49.6	28.8	4.9	0
U2 - cycle-1 oracle	2.2	12.6	43.8	32.9	8.5	0
U3 - shared alpha	2.2	13.8	49.8	29.7	4.5	0
U3 - dose-specific alpha	2.2	14.1	49.4	29.7	4.6	0
U3 - cycle-1 oracle	2.2	12.5	44.6	32.6	8.1	0
OBD selection (%)						
U2 - shared alpha	8	49.8	29.1	9.7	0.9	2.5
U2 - dose-specific alpha	8.4	48.8	30.1	8.9	1.3	2.5
U2 - cycle-1 oracle	9.3	48.6	26.9	11	1.8	2.4
U3 - shared alpha	10.2	53	25.3	8.2	0.7	2.6
U3 - dose-specific alpha	10.1	53.8	24.6	8.2	0.8	2.5
U3 - cycle-1 oracle	11.9	50.4	24.3	9.5	1.3	2.6
Mean administrations						
U2 - shared alpha	4.9	10.6	8.2	4.6	1.4	-
U2 - dose-specific alpha	4.8	10.5	8.3	4.6	1.4	-
U2 - cycle-1 oracle	4.7	10.2	8.2	4.9	1.7	-
U3 - shared alpha	5.1	11.1	7.8	4.4	1.3	-
U3 - dose-specific alpha	5.1	11.3	7.7	4.3	1.3	-
U3 - cycle-1 oracle	4.9	10.6	7.9	4.7	1.6	-
Mean IPDE administrations						
U2 - shared alpha	0.7	5	2.6	1.7	0.5	-
U2 - dose-specific alpha	0.7	5	2.6	1.7	0.5	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.8	5.2	2.5	1.6	0.4	-
U3 - dose-specific alpha	0.9	5.3	2.4	1.6	0.4	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1052.6
U2 - dose-specific alpha						1052.2
U2 - cycle-1 oracle						1646.6
U3 - shared alpha						1048.1
U3 - dose-specific alpha						1046.8
U3 - cycle-1 oracle						1645.9

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
MTD selection (%)						
U2 - shared alpha	54.3	2.3	0	0	0	43.4
U2 - dose-specific alpha	54.4	2.3	0	0	0	43.3
U2 - cycle-1 oracle	56.4	4.2	0	0	0	39.4
U3 - shared alpha	54.8	2.2	0	0	0	43
U3 - dose-specific alpha	55.2	2.1	0	0	0	42.7
U3 - cycle-1 oracle	57	4.3	0	0	0	38.7
OBD selection (%)						
U2 - shared alpha	52.5	1	0	0	0	46.5
U2 - dose-specific alpha	52	1.1	0	0	0	46.9
U2 - cycle-1 oracle	57.2	1.7	0	0	0	41.1
U3 - shared alpha	52.9	1	0	0	0	46.1
U3 - dose-specific alpha	52.6	1	0	0	0	46.4
U3 - cycle-1 oracle	57	1.8	0	0	0	41.2
Mean administrations						
U2 - shared alpha	19.8	2.3	0.1	0	0	-
U2 - dose-specific alpha	19.8	2.3	0.1	0	0	-
U2 - cycle-1 oracle	20.2	2.5	0.2	0	0	-
U3 - shared alpha	20	2.2	0.1	0	0	-
U3 - dose-specific alpha	19.9	2.3	0.1	0	0	-
U3 - cycle-1 oracle	20.3	2.4	0.2	0	0	-
Mean IPDE administrations						
U2 - shared alpha	3.8	0.9	0	0	0	-
U2 - dose-specific alpha	3.8	0.9	0	0	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	3.8	0.9	0	0	0	-
U3 - dose-specific alpha	3.8	0.9	0	0	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						968.5
U2 - cycle-1 oracle						966.7
U3 - shared alpha						1273.1
U3 - dose-specific alpha						968.8
U3 - cycle-1 oracle						967.9
						1271.4
Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
MTD selection (%)						
U2 - shared alpha	14.1	53.7	30.3	1.9	0	0
U2 - dose-specific alpha	13.7	54.9	29.2	2.2	0	0
U2 - cycle-1 oracle	15	50.9	30.5	3.6	0	0
U3 - shared alpha	14	54.5	29.3	2.2	0	0
U3 - dose-specific alpha	14.2	52.7	30.7	2.4	0	0
U3 - cycle-1 oracle	14.8	48.8	32.2	4.2	0	0
OBD selection (%)						
U2 - shared alpha	15	60.2	12.3	1.3	0	11.2
U2 - dose-specific alpha	14.6	60.8	12	1.4	0	11.2
U2 - cycle-1 oracle	14.8	58.1	11.1	2.5	0	13.5
U3 - shared alpha	16.3	61.4	9.9	1.3	0	11.1
U3 - dose-specific alpha	16.1	60.7	9.7	1.5	0	12
U3 - cycle-1 oracle	16	58.2	9.7	2.5	0	13.6
Mean administrations						
U2 - shared alpha	8.4	13.3	5.3	1.8	0.1	-
U2 - dose-specific alpha	8.4	13.3	5.4	1.8	0.1	-
U2 - cycle-1 oracle	8.1	12.8	5.5	2.1	0.2	-
U3 - shared alpha	8.7	13.4	5	1.7	0.1	-
U3 - dose-specific alpha	8.7	13.4	5	1.7	0.1	-
U3 - cycle-1 oracle	8.4	12.9	5.3	2	0.2	-
Mean IPDE administrations						
U2 - shared alpha	2	5.9	1.5	0.7	0	-
U2 - dose-specific alpha	2	5.9	1.6	0.6	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.1	5.9	1.5	0.6	0	-
U3 - dose-specific alpha	2.1	5.9	1.5	0.6	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						1038.6
U2 - cycle-1 oracle						1039.7
U3 - shared alpha						1599.2
U3 - dose-specific alpha						1034.5
U3 - cycle-1 oracle						1033.6
						1598.6

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
MTD selection (%)						
U2 - shared alpha	73.8	2.7	0	0	0	23.5
U2 - dose-specific alpha	74.3	2.6	0	0	0	23.1
U2 - cycle-1 oracle	75.3	4.7	0.1	0	0	19.9
U3 - shared alpha	72.9	3.3	0.1	0	0	23.7
U3 - dose-specific alpha	73.9	2.8	0.1	0	0	23.2
U3 - cycle-1 oracle	75.4	5	0.1	0	0	19.5
OBD selection (%)						
U2 - shared alpha	7.5	0.6	0	0	0	91.9
U2 - dose-specific alpha	7.4	0.6	0	0	0	92
U2 - cycle-1 oracle	7.8	1.3	0	0	0	90.9
U3 - shared alpha	7.5	1.1	0	0	0	91.4
U3 - dose-specific alpha	7.2	0.8	0	0	0	92
U3 - cycle-1 oracle	7.7	1.6	0.1	0	0	90.6
Mean administrations						
U2 - shared alpha	11.1	3.8	0.5	0	0	-
U2 - dose-specific alpha	11.1	3.9	0.5	0	0	-
U2 - cycle-1 oracle	11	4.2	0.7	0.1	0	-
U3 - shared alpha	11.1	3.8	0.5	0	0	-
U3 - dose-specific alpha	11.1	3.8	0.5	0.1	0	-
U3 - cycle-1 oracle	11	4.2	0.7	0.1	0	-
Mean IPDE administrations						
U2 - shared alpha	2.9	2.1	0.2	0	0	-
U2 - dose-specific alpha	2.9	2.1	0.2	0	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.9	2.1	0.2	0	0	-
U3 - dose-specific alpha	2.9	2.1	0.2	0	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						557.9
U2 - cycle-1 oracle						559
U3 - shared alpha						882.6
U3 - dose-specific alpha						557.2
U3 - cycle-1 oracle						558.3
						878.5

Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
MTD selection (%)						
U2 - shared alpha	0	0.1	0.1	37	62.8	0
U2 - dose-specific alpha	0	0.1	0.1	36.1	63.7	0
U2 - cycle-1 oracle	0	0.2	0	30.5	69.3	0
U3 - shared alpha	0	0.1	0.2	40.9	58.8	0
U3 - dose-specific alpha	0	0.1	0.2	41.5	58.2	0
U3 - cycle-1 oracle	0	0.2	0	33.9	65.9	0
OBD selection (%)						
U2 - shared alpha	1.5	1	3	59.1	35.4	0
U2 - dose-specific alpha	1.5	0.9	2.5	59.2	35.9	0
U2 - cycle-1 oracle	2.3	1.9	2.9	56.3	36.5	0.1
U3 - shared alpha	2.7	1.9	2.7	62.6	30.1	0
U3 - dose-specific alpha	2.6	2.1	4	62	29.3	0
U3 - cycle-1 oracle	4	3.3	4.2	57.8	30.6	0.1
Mean administrations						
U2 - shared alpha	3.1	3.1	3.8	13	6.9	-
U2 - dose-specific alpha	3.1	3.2	3.8	13	7	-
U2 - cycle-1 oracle	3.1	3.2	4	12.8	6.9	-
U3 - shared alpha	3.1	3.2	3.9	13.6	6.2	-
U3 - dose-specific alpha	3.1	3.2	3.9	13.6	6.1	-
U3 - cycle-1 oracle	3.1	3.2	4.1	13.2	6.4	-
Mean IPDE administrations						
U2 - shared alpha	0	2.4	1.2	4.5	2.1	-
U2 - dose-specific alpha	0	2.4	1.2	4.5	2.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0	2.4	1.2	4.7	1.8	-
U3 - dose-specific alpha	0.1	2.4	1.2	4.7	1.8	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						1090.8
U2 - cycle-1 oracle						1090.6
U3 - shared alpha						1663.2
U3 - dose-specific alpha						1088.3
U3 - cycle-1 oracle						1087.5
						1663.2

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
MTD selection (%)						
U2 - shared alpha	3.2	37	52.2	7.5	0.1	0
U2 - dose-specific alpha	3.1	36.8	52.7	7.3	0.1	0
U2 - cycle-1 oracle	2.6	35.4	51.7	10	0.3	0
U3 - shared alpha	2.7	36.7	52.7	7.8	0.1	0
U3 - dose-specific alpha	3.3	36.6	52.2	7.8	0.1	0
U3 - cycle-1 oracle	2.2	36.1	51.2	10.2	0.3	0
OBD selection (%)						
U2 - shared alpha	28.2	33.2	33	4.1	0.1	1.4
U2 - dose-specific alpha	28.6	33	33.6	3.3	0.1	1.4
U2 - cycle-1 oracle	28	31.4	33.6	5.5	0	1.5
U3 - shared alpha	32.2	34.2	29.5	2.8	0	1.3
U3 - dose-specific alpha	33	33.7	29.1	2.9	0	1.3
U3 - cycle-1 oracle	32.7	31.9	28.7	5.3	0	1.4
Mean administrations						
U2 - shared alpha	8.8	9.9	8.5	2.4	0.2	-
U2 - dose-specific alpha	9	9.8	8.5	2.4	0.2	-
U2 - cycle-1 oracle	8.9	9.3	8.7	2.7	0.4	-
U3 - shared alpha	9.4	10	8.2	2.1	0.2	-
U3 - dose-specific alpha	9.4	10	8.1	2.1	0.2	-
U3 - cycle-1 oracle	9.5	9.3	8.3	2.5	0.3	-
Mean IPDE administrations						
U2 - shared alpha	2.2	4.3	2.4	0.8	0.1	-
U2 - dose-specific alpha	2.3	4.2	2.4	0.8	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.5	4.3	2.2	0.8	0.1	-
U3 - dose-specific alpha	2.5	4.3	2.2	0.8	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1109.8
U2 - dose-specific alpha						1107.9
U2 - cycle-1 oracle						1654.6
U3 - shared alpha						1103.1
U3 - dose-specific alpha						1104.5
U3 - cycle-1 oracle						1656.4
Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
MTD selection (%)						
U2 - shared alpha	14.6	36.9	38.8	9.1	0.3	0.3
U2 - dose-specific alpha	15	36	38.7	9.7	0.3	0.3
U2 - cycle-1 oracle	13.9	32.7	41.1	11.8	0.2	0.3
U3 - shared alpha	14.5	36.1	39.9	8.9	0.3	0.3
U3 - dose-specific alpha	13.7	37.4	39.3	9	0.3	0.3
U3 - cycle-1 oracle	13.5	33.2	41.1	11.7	0.2	0.3
OBD selection (%)						
U2 - shared alpha	20.6	33.2	28.6	5.2	0.3	12.1
U2 - dose-specific alpha	19.8	33.5	28.9	5.1	0.2	12.5
U2 - cycle-1 oracle	20.2	29.4	31.5	6.1	0.1	12.7
U3 - shared alpha	23.7	33.1	26.7	4.1	0.2	12.2
U3 - dose-specific alpha	23.1	35.2	25.6	4	0.3	11.8
U3 - cycle-1 oracle	21.3	33.6	26.7	5.7	0.1	12.6
Mean administrations						
U2 - shared alpha	8.7	9.2	7.7	2.7	0.3	-
U2 - dose-specific alpha	8.6	9.2	7.7	2.6	0.3	-
U2 - cycle-1 oracle	8.2	8.7	8.1	3	0.4	-
U3 - shared alpha	9.1	9.4	7.5	2.4	0.2	-
U3 - dose-specific alpha	9	9.3	7.5	2.4	0.2	-
U3 - cycle-1 oracle	8.5	8.8	7.9	2.8	0.4	-
Mean IPDE administrations						
U2 - shared alpha	2.2	4.3	2.2	1	0.1	-
U2 - dose-specific alpha	2.1	4.4	2.2	1	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.3	4.4	2.1	0.9	0.1	-
U3 - dose-specific alpha	2.3	4.4	2.1	0.9	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1030
U2 - dose-specific alpha						1025.8
U2 - cycle-1 oracle						1578.8
U3 - shared alpha						1024.5
U3 - dose-specific alpha						1024.9
U3 - cycle-1 oracle						1578.7

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
MTD selection (%)						
U2 - shared alpha	4.9	23	54.5	16.5	1.1	0
U2 - dose-specific alpha	4.6	23.7	53.3	17.5	0.9	0
U2 - cycle-1 oracle	5.2	20.4	52.3	19.8	2.3	0
U3 - shared alpha	4.6	24.7	53.4	16.6	0.7	0
U3 - dose-specific alpha	4.5	25.4	52.4	16.9	0.8	0
U3 - cycle-1 oracle	5.2	20.5	52.8	19.5	2	0
OBD selection (%)						
U2 - shared alpha	14.4	25.7	45.5	9.3	0.4	4.7
U2 - dose-specific alpha	14.3	24.5	46.2	10	0.3	4.7
U2 - cycle-1 oracle	13.3	24.6	45.4	10.8	0.4	5.5
U3 - shared alpha	17.4	27.7	42.3	7.5	0.2	4.9
U3 - dose-specific alpha	17.8	27.6	42.1	7.6	0.3	4.6
U3 - cycle-1 oracle	16.4	26.4	42.2	8.9	0.5	5.6
Mean administrations						
U2 - shared alpha	6.7	8.6	10	3.5	0.6	-
U2 - dose-specific alpha	6.7	8.6	10	3.5	0.6	-
U2 - cycle-1 oracle	6.3	8.2	10.1	3.9	0.8	-
U3 - shared alpha	7	8.8	9.8	3.2	0.5	-
U3 - dose-specific alpha	7	8.8	9.9	3.2	0.5	-
U3 - cycle-1 oracle	6.6	8.3	10.1	3.7	0.7	-
Mean IPDE administrations						
U2 - shared alpha	1.4	4.1	2.9	1.2	0.2	-
U2 - dose-specific alpha	1.5	4.1	2.9	1.2	0.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.6	4.2	2.9	1.1	0.2	-
U3 - dose-specific alpha	1.6	4.2	2.9	1.1	0.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1077.5
U2 - dose-specific alpha						1077.2
U2 - cycle-1 oracle						1630.8
U3 - shared alpha						1071.9
U3 - dose-specific alpha						1072
U3 - cycle-1 oracle						1629.8
Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
MTD selection (%)						
U2 - shared alpha	48.9	37.8	10.7	0.8	0	1.8
U2 - dose-specific alpha	48.6	37.1	11.6	0.7	0	2
U2 - cycle-1 oracle	45.1	37.6	13.7	1.7	0	1.9
U3 - shared alpha	49.3	36.8	11.5	0.5	0	1.9
U3 - dose-specific alpha	49.2	37	11.2	0.7	0	1.9
U3 - cycle-1 oracle	45.2	36.6	15.2	1.3	0	1.7
OBD selection (%)						
U2 - shared alpha	11.2	39.2	5.4	0.1	0	44.1
U2 - dose-specific alpha	10.6	38.9	6.1	0.1	0	44.3
U2 - cycle-1 oracle	10.1	41.5	6.8	0.2	0	41.4
U3 - shared alpha	11.6	38.7	5.1	0	0	44.6
U3 - dose-specific alpha	11.9	39.2	4.4	0	0	44.5
U3 - cycle-1 oracle	10.2	42.1	6.5	0.2	0	41
Mean administrations						
U2 - shared alpha	9.4	10.7	3.3	0.8	0.1	-
U2 - dose-specific alpha	9.4	10.6	3.3	0.9	0.1	-
U2 - cycle-1 oracle	8.8	10.9	3.9	1.1	0.1	-
U3 - shared alpha	9.6	10.8	3.2	0.8	0.1	-
U3 - dose-specific alpha	9.7	10.6	3.2	0.8	0.1	-
U3 - cycle-1 oracle	9	11.1	3.7	1	0.1	-
Mean IPDE administrations						
U2 - shared alpha	2.3	4.7	1	0.3	0	-
U2 - dose-specific alpha	2.2	4.6	1	0.3	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.3	4.7	0.9	0.3	0	-
U3 - dose-specific alpha	2.4	4.6	1	0.3	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						880.4
U2 - dose-specific alpha						881.3
U2 - cycle-1 oracle						1379.9
U3 - shared alpha						883.8
U3 - dose-specific alpha						878.5
U3 - cycle-1 oracle						1378.2

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
MTD selection (%)						
U2 - shared alpha	31.5	57.4	8.5	0.3	0	2.3
U2 - dose-specific alpha	31.3	57.5	8.8	0.3	0	2.1
U2 - cycle-1 oracle	27.1	59.9	10	0.8	0	2.2
U3 - shared alpha	31.8	58.5	7.2	0.2	0	2.3
U3 - dose-specific alpha	32.8	57.8	7.1	0.2	0	2.1
U3 - cycle-1 oracle	27.7	60	9.4	0.9	0	2
OBD selection (%)						
U2 - shared alpha	63.7	31.4	2.1	0.1	0	2.7
U2 - dose-specific alpha	64.5	30.1	2.9	0.1	0	2.4
U2 - cycle-1 oracle	64.4	29.9	3.2	0.2	0	2.3
U3 - shared alpha	67.5	28.1	1.8	0.1	0	2.5
U3 - dose-specific alpha	68.3	27.5	1.7	0.1	0	2.4
U3 - cycle-1 oracle	67.3	27.6	2.3	0.2	0	2.6
Mean administrations						
U2 - shared alpha	18.1	9	2	0.4	0	-
U2 - dose-specific alpha	18.1	8.9	2.1	0.4	0	-
U2 - cycle-1 oracle	17.7	8.8	2.5	0.5	0	-
U3 - shared alpha	19	8.6	1.6	0.3	0	-
U3 - dose-specific alpha	19	8.6	1.7	0.3	0	-
U3 - cycle-1 oracle	18.5	8.4	2.1	0.4	0	-
Mean IPDE administrations						
U2 - shared alpha	4.4	3.2	0.6	0.1	0	-
U2 - dose-specific alpha	4.4	3.2	0.6	0.1	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	4.8	3.1	0.5	0.1	0	-
U3 - dose-specific alpha	4.8	3.1	0.5	0.1	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1166.3
U2 - dose-specific alpha						1167.6
U2 - cycle-1 oracle						1641.4
U3 - shared alpha						1164.5
U3 - dose-specific alpha						1164.8
U3 - cycle-1 oracle						1638

Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
MTD selection (%)						
U2 - shared alpha	0.4	1.7	17.3	58.7	21.9	0
U2 - dose-specific alpha	0.4	1.8	17	59.1	21.7	0
U2 - cycle-1 oracle	0.5	1.3	14.3	56.9	27	0
U3 - shared alpha	0.4	2.2	17.6	59.5	20.3	0
U3 - dose-specific alpha	0.4	2	16.6	60.6	20.4	0
U3 - cycle-1 oracle	0.5	1.5	14.3	57.8	25.9	0
OBD selection (%)						
U2 - shared alpha	16.3	10.8	15.6	46	10.9	0.4
U2 - dose-specific alpha	15.7	11.7	15.4	45.6	11.2	0.4
U2 - cycle-1 oracle	16.3	12.8	12.7	45.3	12.4	0.5
U3 - shared alpha	23.5	14.2	12.9	39.7	9.2	0.5
U3 - dose-specific alpha	22.8	16.3	12.4	39	9	0.5
U3 - cycle-1 oracle	24.2	15.9	11.6	38.2	9.6	0.5
Mean administrations						
U2 - shared alpha	4.4	5.2	6.4	10.1	3.8	-
U2 - dose-specific alpha	4.4	5.2	6.4	10.2	3.7	-
U2 - cycle-1 oracle	4.3	5	6.1	10.2	4.3	-
U3 - shared alpha	4.7	5.5	6.4	10	3.4	-
U3 - dose-specific alpha	4.7	5.4	6.4	10	3.4	-
U3 - cycle-1 oracle	4.4	5.1	6.1	10.3	3.9	-
Mean IPDE administrations						
U2 - shared alpha	0.6	3.1	2	3.3	1.1	-
U2 - dose-specific alpha	0.6	3.1	2	3.4	1.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.7	3.2	2	3.3	1	-
U3 - dose-specific alpha	0.7	3.2	2	3.3	1.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1085.8
U2 - dose-specific alpha						1087.1
U2 - cycle-1 oracle						1659.7
U3 - shared alpha						1081.6
U3 - dose-specific alpha						1082.3
U3 - cycle-1 oracle						1659.7

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
MTD selection (%)						
U2 - shared alpha	0	30.2	28.1	30	11.7	0
U2 - dose-specific alpha	0	30	29.6	28.9	11.5	0
U2 - cycle-1 oracle	0	25.3	25.9	31.1	17.7	0
U3 - shared alpha	0	35.7	28.6	27.6	8.1	0
U3 - dose-specific alpha	0	34.8	28.7	28.1	8.4	0
U3 - cycle-1 oracle	0	30.1	25.5	29.9	14.5	0
OBD selection (%)						
U2 - shared alpha	2.9	38.1	31.3	23.2	4.5	0
U2 - dose-specific alpha	2.7	39.2	30.4	23.3	4.4	0
U2 - cycle-1 oracle	4.6	35.3	29.1	25.3	5.7	0
U3 - shared alpha	4.8	44.8	28.3	20	2.1	0
U3 - dose-specific alpha	4.9	42.6	29.4	20.6	2.5	0
U3 - cycle-1 oracle	6.4	42.4	27.4	19.6	4.2	0
Mean administrations						
U2 - shared alpha	3.8	12.5	7.6	4.8	1.2	-
U2 - dose-specific alpha	3.8	12.6	7.7	4.7	1.2	-
U2 - cycle-1 oracle	3.8	11.6	7.4	5.5	1.7	-
U3 - shared alpha	4	13.6	7.3	4.3	0.8	-
U3 - dose-specific alpha	4	13.3	7.4	4.5	0.8	-
U3 - cycle-1 oracle	3.8	12.4	7.2	5.1	1.4	-
Mean IPDE administrations						
U2 - shared alpha	0.3	4.9	2.3	1.4	0.4	-
U2 - dose-specific alpha	0.3	4.9	2.3	1.4	0.4	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.3	5.2	2.2	1.3	0.3	-
U3 - dose-specific alpha	0.3	5.2	2.2	1.3	0.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1145.8
U2 - dose-specific alpha						1144.5
U2 - cycle-1 oracle						1664.1
U3 - shared alpha						1140.5
U3 - dose-specific alpha						1140.8
U3 - cycle-1 oracle						1664.1
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
MTD selection (%)						
U2 - shared alpha	0.1	0.3	7.6	42.8	49.2	0
U2 - dose-specific alpha	0.1	0.4	8	42.9	48.6	0
U2 - cycle-1 oracle	0.1	0.3	7.3	34.8	57.5	0
U3 - shared alpha	0.1	0.3	8.7	46.9	44	0
U3 - dose-specific alpha	0.1	0.2	8.1	48.4	43.2	0
U3 - cycle-1 oracle	0.1	0.3	7	40.9	51.7	0
OBD selection (%)						
U2 - shared alpha	11.4	14	35	29.1	10.4	0.1
U2 - dose-specific alpha	11.9	13.4	34.9	29.2	10.4	0.2
U2 - cycle-1 oracle	12.9	14.7	33.6	26.3	12.3	0.2
U3 - shared alpha	16.5	18.3	33.6	25.8	5.7	0.1
U3 - dose-specific alpha	16.7	18	32.7	27.4	5.1	0.1
U3 - cycle-1 oracle	18.6	16.8	33.6	22.6	8.2	0.2
Mean administrations						
U2 - shared alpha	3.5	4.3	8.7	8.7	4.7	-
U2 - dose-specific alpha	3.5	4.3	8.7	8.7	4.7	-
U2 - cycle-1 oracle	3.5	4.4	8.6	8.5	5	-
U3 - shared alpha	3.6	4.6	9.1	8.6	4	-
U3 - dose-specific alpha	3.6	4.7	9	8.6	4.1	-
U3 - cycle-1 oracle	3.5	4.6	8.9	8.4	4.5	-
Mean IPDE administrations						
U2 - shared alpha	0.2	2.9	2.9	3.3	1.5	-
U2 - dose-specific alpha	0.2	2.9	2.9	3.3	1.5	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.3	3	3	3.2	1.3	-
U3 - dose-specific alpha	0.3	3	3	3.2	1.3	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1055.9
U2 - dose-specific alpha						1055.2
U2 - cycle-1 oracle						1662.7
U3 - shared alpha						1052.1
U3 - dose-specific alpha						1051.7
U3 - cycle-1 oracle						1662.7

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
MTD selection (%)						
U2 - shared alpha	0	0.3	7.9	52	39.8	0
U2 - dose-specific alpha	0	0.1	7.4	52.9	39.6	0
U2 - cycle-1 oracle	0	0.3	7.2	45.7	46.8	0
U3 - shared alpha	0	0.2	7.1	54.9	37.8	0
U3 - dose-specific alpha	0	0.1	7.6	55.1	37.2	0
U3 - cycle-1 oracle	0	0.3	6.6	49	44.1	0
OBD selection (%)						
U2 - shared alpha	8.8	5.9	11.6	47.3	26.3	0.1
U2 - dose-specific alpha	8.4	5.8	11.4	47.7	26.6	0.1
U2 - cycle-1 oracle	8.4	5.6	11.2	44.5	30.1	0.2
U3 - shared alpha	12.2	8.2	12	46.1	21.4	0.1
U3 - dose-specific alpha	12.6	8.4	11.7	45.7	21.5	0.1
U3 - cycle-1 oracle	12.4	8.6	11.3	43.9	23.6	0.2
Mean administrations						
U2 - shared alpha	3.4	3.7	6	10.7	6.2	-
U2 - dose-specific alpha	3.3	3.8	6.1	10.7	6.2	-
U2 - cycle-1 oracle	3.3	3.7	6.1	10.5	6.4	-
U3 - shared alpha	3.5	3.8	6.2	10.9	5.7	-
U3 - dose-specific alpha	3.5	3.9	6.1	10.9	5.6	-
U3 - cycle-1 oracle	3.4	3.8	6.2	10.6	5.9	-
Mean IPDE administrations						
U2 - shared alpha	0.2	2.7	1.8	3.8	1.7	-
U2 - dose-specific alpha	0.1	2.7	1.8	3.8	1.7	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.2	2.8	1.9	3.8	1.5	-
U3 - dose-specific alpha	0.2	2.8	1.8	3.9	1.5	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1090.3
U2 - dose-specific alpha						1089.2
U2 - cycle-1 oracle						1663.1
U3 - shared alpha						1087
U3 - dose-specific alpha						1086.8
U3 - cycle-1 oracle						1662.8

Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
MTD selection (%)						
U2 - shared alpha	1.4	11	44	33.6	10	0
U2 - dose-specific alpha	1.4	10.8	44	33.6	10.2	0
U2 - cycle-1 oracle	1	10.7	36.8	37.8	13.7	0
U3 - shared alpha	1.5	10.7	43.4	35.7	8.7	0
U3 - dose-specific alpha	1.4	10.4	44	35.5	8.7	0
U3 - cycle-1 oracle	0.9	10.3	37.5	39.3	12	0
OBD selection (%)						
U2 - shared alpha	12.7	13.2	47.5	19.2	3.1	4.3
U2 - dose-specific alpha	13.2	12.9	47.7	19	3.2	4
U2 - cycle-1 oracle	12.4	13.2	44	21.9	4.5	4
U3 - shared alpha	15.8	16	44.1	17.9	2.2	4
U3 - dose-specific alpha	16.2	15.9	44.2	17.6	2	4.1
U3 - cycle-1 oracle	16.1	15.8	42.5	19.3	2.4	3.9
Mean administrations						
U2 - shared alpha	5.1	6.7	9.9	5.9	2.1	-
U2 - dose-specific alpha	5.1	6.6	9.9	5.9	2.1	-
U2 - cycle-1 oracle	5	6.3	9.7	6.2	2.5	-
U3 - shared alpha	5.5	6.8	9.9	5.8	1.7	-
U3 - dose-specific alpha	5.5	6.8	9.9	5.7	1.9	-
U3 - cycle-1 oracle	5.2	6.5	9.7	6	2.2	-
Mean IPDE administrations						
U2 - shared alpha	0.9	3.7	3.1	2.1	0.7	-
U2 - dose-specific alpha	0.9	3.7	3.1	2.1	0.7	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.1	3.8	3.1	2	0.6	-
U3 - dose-specific alpha	1.1	3.8	3.1	2	0.6	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1054
U2 - dose-specific alpha						1053.7
U2 - cycle-1 oracle						1646.2
U3 - shared alpha						1050.1
U3 - dose-specific alpha						1052
U3 - cycle-1 oracle						1647.5

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
MTD selection (%)						
U2 - shared alpha	0.1	23.7	48.9	23	4.3	0
U2 - dose-specific alpha	0.1	23.5	47.8	23.4	5.2	0
U2 - cycle-1 oracle	0.1	19.9	43.9	28.2	7.9	0
U3 - shared alpha	0.1	28.2	46.3	21.7	3.7	0
U3 - dose-specific alpha	0.1	27.4	47.9	20.6	4	0
U3 - cycle-1 oracle	0.1	23.6	44	25.4	6.9	0
OBD selection (%)						
U2 - shared alpha	7.3	43.9	35.2	11.5	2	0.1
U2 - dose-specific alpha	7.7	42.1	34.9	12.7	2.5	0.1
U2 - cycle-1 oracle	8.5	41.7	31.4	14.9	3.4	0.1
U3 - shared alpha	11	46.5	30.8	9.7	1.9	0.1
U3 - dose-specific alpha	11.1	47.7	31	8.4	1.7	0.1
U3 - cycle-1 oracle	12.6	46.1	28.5	10.4	2.3	0.1
Mean administrations						
U2 - shared alpha	4.4	12.3	8.6	3.7	0.9	-
U2 - dose-specific alpha	4.4	12.1	8.6	3.8	1	-
U2 - cycle-1 oracle	4.4	11.5	8.5	4.2	1.4	-
U3 - shared alpha	4.6	13.2	8	3.4	0.8	-
U3 - dose-specific alpha	4.7	13.1	8	3.3	0.8	-
U3 - cycle-1 oracle	4.6	12.4	8	3.8	1.2	-
Mean IPDE administrations						
U2 - shared alpha	0.5	4.8	2.4	1.2	0.3	-
U2 - dose-specific alpha	0.5	4.8	2.4	1.2	0.3	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.6	5.2	2.2	1	0.2	-
U3 - dose-specific alpha	0.6	5.2	2.2	1	0.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1149
U2 - dose-specific alpha						1148.4
U2 - cycle-1 oracle						1663.2
U3 - shared alpha						1142.2
U3 - dose-specific alpha						1142.4
U3 - cycle-1 oracle						1663.2
Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
MTD selection (%)						
U2 - shared alpha	24.9	34.4	27	10.4	3.1	0.2
U2 - dose-specific alpha	24.8	33.9	27.7	10.5	2.9	0.2
U2 - cycle-1 oracle	21.8	33.6	26.5	12.8	5.1	0.2
U3 - shared alpha	23.9	36.4	26.7	10	2.8	0.2
U3 - dose-specific alpha	24.2	36.1	26.4	10.2	2.9	0.2
U3 - cycle-1 oracle	22.5	32.7	26.7	13.6	4.3	0.2
OBD selection (%)						
U2 - shared alpha	13.4	41.4	15.4	5.1	1	23.7
U2 - dose-specific alpha	14.3	40.6	16.4	4.6	1	23.1
U2 - cycle-1 oracle	12.7	42	13.9	6.8	1.5	23.1
U3 - shared alpha	16.8	42.4	13.5	4	0.5	22.8
U3 - dose-specific alpha	15.8	42.6	13.3	4.5	0.6	23.2
U3 - cycle-1 oracle	16.3	42	10.4	6.5	1.2	23.6
Mean administrations						
U2 - shared alpha	8.1	10.2	5.3	2.9	0.8	-
U2 - dose-specific alpha	8.1	10.2	5.3	2.9	0.8	-
U2 - cycle-1 oracle	7.7	9.9	5.4	3.3	1.1	-
U3 - shared alpha	8.4	10.3	5.1	2.7	0.7	-
U3 - dose-specific alpha	8.4	10.4	5.1	2.7	0.7	-
U3 - cycle-1 oracle	7.9	10	5.4	3.2	1	-
Mean IPDE administrations						
U2 - shared alpha	1.9	4.9	1.7	1.1	0.3	-
U2 - dose-specific alpha	1.8	4.9	1.7	1.1	0.3	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2	5	1.6	1	0.2	-
U3 - dose-specific alpha	2	5	1.6	1	0.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						958.5
U2 - dose-specific alpha						959.2
U2 - cycle-1 oracle						1521.3
U3 - shared alpha						957
U3 - dose-specific alpha						955.6
U3 - cycle-1 oracle						1523.4

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
MTD selection (%)						
U2 - shared alpha	19.8	43.4	28.9	6.5	1.4	0
U2 - dose-specific alpha	19.5	43.9	28.8	6.5	1.3	0
U2 - cycle-1 oracle	19.6	39.3	30.1	8.8	2.2	0
U3 - shared alpha	18.9	43.6	29	7.2	1.3	0
U3 - dose-specific alpha	19	43.8	28.9	7.1	1.2	0
U3 - cycle-1 oracle	19.4	38.7	30.5	9.4	2	0
OBD selection (%)						
U2 - shared alpha	13.4	50.2	13.7	2.5	0.4	19.8
U2 - dose-specific alpha	13	50.2	13.5	2.8	0.4	20.1
U2 - cycle-1 oracle	14.5	47.1	13.2	3.3	0.6	21.3
U3 - shared alpha	16.1	50.3	11.6	2.4	0.2	19.4
U3 - dose-specific alpha	16.1	50.9	11.3	2	0.2	19.5
U3 - cycle-1 oracle	16.7	48	11.4	2.3	0.6	21
Mean administrations						
U2 - shared alpha	7.9	11.6	5.4	2.5	0.7	-
U2 - dose-specific alpha	8	11.5	5.3	2.5	0.7	-
U2 - cycle-1 oracle	7.7	11.2	5.6	2.9	0.8	-
U3 - shared alpha	8.3	11.5	5.2	2.4	0.6	-
U3 - dose-specific alpha	8.3	11.5	5.2	2.4	0.6	-
U3 - cycle-1 oracle	7.9	11.2	5.4	2.8	0.8	-
Mean IPDE administrations						
U2 - shared alpha	1.8	5.5	1.7	1	0.2	-
U2 - dose-specific alpha	1.8	5.5	1.7	1	0.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2	5.5	1.6	0.9	0.2	-
U3 - dose-specific alpha	2	5.5	1.6	0.9	0.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						979.6
U2 - cycle-1 oracle						977.4
U3 - shared alpha						1557.9
U3 - dose-specific alpha						975.8
U3 - cycle-1 oracle						976.4
						1558.6